

IMO 2026

Complete Solutions

Claude Fable 5

Independently graded: **42/42**

Autonomous run, no human intervention.

Solutions as written by the model; all six problems.

github.com/deedy/imo-2026

IMO 2026 — Problem 1

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Approaches tried

- **Per-prime exponent dynamics + monovariant + gcd invariant** (see `approaches/potential-and-invariant`) — **successful, used in the final proof.** A move sends the exponent pair $(a, b) = (v_p(m), v_p(n))$ to $(\min(a, b), \max(a, b) - \min(a, b))$. Termination via the strictly decreasing potential $T + N$ ($T = \sum_i \Omega(x_i)$, $N = \text{number of entries} > 1$); part (b) via the invariance of $g_p = \gcd_i v_p(x_i)$ for each prime p .

- Computational verification (`code/check.py`): exhaustive BFS over the whole game tree for all multisets $\{a, b, c\} \subseteq [2, 14]$, 300 random 4-tuples in $[2, 30]$, 60 random 5-tuples in $[2, 12]$. Confirmed: every terminal state has exactly one entry > 1 , the surviving value is identical across all terminal states, and it equals $\prod_p p^{\gcd_i v_p(a_i)}$; also verified $T + N$ strictly decreases along every edge. All checks passed.

Current best

Complete proof of both parts. Explicit formula: if the initial numbers are $a_1, \dots, a_{2026}$, then the process always terminates and the unique surviving integer greater than 1 is

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))},$$

which depends only on the initial numbers — proving (b). Part (a) follows from the strictly decreasing potential $T + N$ (termination) plus the fact that each move outputs at least one integer > 1 (so the count of entries > 1 never drops below 1).

Full proof

Throughout, the **state** of the blackboard is the tuple $(x_1, \dots, x_{2026})$ of the written numbers. For a prime p and a positive integer x , $v_p(x)$ denotes the exponent of p in x , and $\Omega(x) = \sum_p v_p(x)$ is the number of prime factors of x counted with multiplicity (so $\Omega(1) = 0$, and $\Omega(x) \geq 1$ for $x > 1$). We use the standard facts

$$v_p(\gcd(m, n)) = \min(v_p(m), v_p(n)), \quad v_p(\text{lcm}(m, n)) = \max(v_p(m), v_p(n)).$$

Lemma 1 (effect of a move on exponents)

Let a move replace $m, n > 1$ by $g = \gcd(m, n)$ and $h = \text{lcm}(m, n) / \gcd(m, n)$. Then g, h are positive integers, and for every prime p , writing $a = v_p(m)$, $b = v_p(n)$,

$$v_p(g) = \min(a, b), \quad v_p(h) = \max(a, b) - \min(a, b).$$

Proof. For every prime p we have $v_p(\gcd(m, n)) = \min(a, b) \leq \max(a, b) = v_p(\text{lcm}(m, n))$; hence $\gcd(m, n) \mid \text{lcm}(m, n)$, so h is a positive integer and $v_p(h) = \max(a, b) - \min(a, b)$. ■

All other entries are unchanged by the move, and every entry always remains a positive integer.

Lemma 2 (a move produces at least one entry > 1)

With the notation of Lemma 1, $\max(g, h) > 1$.

Proof. Suppose $h = 1$. Then $\text{lcm}(m, n) = \gcd(m, n)$. Since $\gcd(m, n) \mid m \mid \text{lcm}(m, n)$, this forces $g = \gcd(m, n) = m = \text{lcm}(m, n)$, and $m > 1$, so $g > 1$. Hence $h = 1$ implies $g > 1$; in all cases at least one of g, h exceeds 1. ■

Lemma 3 (monovariant)

For a state $S = (x_1, \dots, x_{2026})$ define

$$T(S) = \sum_{i=1}^{2026} \Omega(x_i), \quad N(S) = \#\{i : x_i > 1\}, \quad \Phi(S) = T(S) + N(S).$$

Then every move strictly decreases Φ , i.e. $\Phi(S') \leq \Phi(S) - 1$ for the state S' after the move.

Proof. Let the move replace m, n by g, h as above. Fix a prime p and let $a = v_p(m), b = v_p(n)$. By Lemma 1,

$$v_p(g) + v_p(h) = \min(a, b) + (\max(a, b) - \min(a, b)) = \max(a, b) = a + b - \min(a, b) = v_p(m) + v_p(n) - v_p(g).$$

Summing over all primes p gives $\Omega(g) + \Omega(h) = \Omega(m) + \Omega(n) - \Omega(g)$, hence

$$T(S') = T(S) - \Omega(g).$$

For N : the two chosen positions contribute 2 to $N(S)$ (both $m, n > 1$) and at most 2 to $N(S')$; all other positions are unchanged. Hence $N(S') \leq N(S)$ always. We distinguish two cases.

Case $g = 1$. Then $h = \text{lcm}(m, n) = mn / \gcd(m, n) = mn > 1$, so the two positions contribute exactly 1 to $N(S')$, giving $N(S') = N(S) - 1$; and $T(S') = T(S) - \Omega(1) = T(S)$. Thus $\Phi(S') = \Phi(S) - 1$.

Case $g > 1$. Then $\Omega(g) \geq 1$, so $T(S') \leq T(S) - 1$, and $N(S') \leq N(S)$. Thus $\Phi(S') \leq \Phi(S) - 1$.

In both cases Φ decreases by at least 1. ■

Part (a)

Termination. Φ is a nonnegative integer for every state (each $x_i \geq 1$), and by Lemma 3 it strictly decreases with every move. Hence at most $\Phi(S_0)$ moves can be made from the initial state S_0 : after finitely many moves the process stops.

The process stops exactly when $N \leq 1$. A move requires two entries > 1 in different places, and any two entries > 1 can be chosen; thus a move is possible if and only if $N \geq 2$. So the final state satisfies $N \leq 1$.

$N \geq 1$ at all times. Initially $N = 2026 \geq 1$. Consider any move, performed from a state with $N \geq 2$. The two chosen positions held two entries > 1 and, by Lemma 2, hold at least one entry > 1 afterwards; all other positions are unchanged. Hence N decreases by at most 1 per move, so after the move $N \geq 2 - 1 = 1$. By induction, $N \geq 1$ in every state ever reached.

Combining: the process terminates after finitely many moves, and in the final state $1 \leq N \leq 1$, i.e. **exactly one integer on the blackboard is greater than 1**. Call it M . This proves (a). ■

Part (b)

For a state $S = (x_1, \dots, x_{2026})$ and a prime p define

$$g_p(S) = \gcd(v_p(x_1), \dots, v_p(x_{2026})) \in \mathbb{Z}_{\geq 0},$$

with the standard conventions $\gcd(t, 0) = t$ and $\gcd(0, 0, \dots, 0) = 0$ (so $g_p(S) = 0$ exactly when p divides no entry).

Lemma 4 (invariance of g_p)

For every prime p and every move $S \rightarrow S'$ we have $g_p(S') = g_p(S)$.

Proof. First, for all integers $a, b \geq 0$,

$$\gcd(\min(a, b), \max(a, b) - \min(a, b)) = \gcd(a, b). \quad (*)$$

Indeed, WLOG $a \leq b$; an integer d divides both a and $b - a$ if and only if it divides both a and b , so the pairs $(a, b - a)$ and (a, b) have the same common divisors and hence the same gcd. (If $a = b = 0$, both sides are 0.)

Now let the move act on entries m, n with $a = v_p(m)$, $b = v_p(n)$, and let r denote the gcd of the values $v_p(x_i)$ over all the *other* 2024 positions. By Lemma 1 and the associativity of gcd,

$$g_p(S') = \gcd\left(r, \gcd(\min(a, b), \max(a, b) - \min(a, b))\right) \stackrel{(*)}{=} \gcd(r, \gcd(a, b)) = g_p(S). \quad \blacksquare$$

Conclusion of (b). Let $S_0 = (a_1, \dots, a_{2026})$ be the initial state and let S_f be any final state reached by any sequence of choices. By part (a), S_f consists of one integer $M > 1$ and 2025 ones. Hence for every prime p ,

$$g_p(S_f) = \gcd(v_p(M), 0, \dots, 0) = v_p(M).$$

By Lemma 4, $g_p(S_f) = g_p(S_0)$ for every prime p . Therefore

$$v_p(M) = g_p(S_0) = \gcd(v_p(a_1), \dots, v_p(a_{2026})) \quad \text{for every prime } p,$$

and since only the finitely many primes dividing $a_1 \cdots a_{2026}$ give a nonzero exponent,

$$M = \prod_{p \text{ prime}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

This expression depends only on the initial numbers $a_1, \dots, a_{2026}$ and not on any of Confucius's choices. Hence the value of M is the same for every legal sequence of moves. $\blacksquare$

(Consistency check: since $a_1 > 1$, some prime p satisfies $v_p(a_1) \geq 1$, whence $g_p(S_0) \geq 1$ and indeed $M > 1$, matching part (a).)

IMO 2026 — Problem 2

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Approaches tried

- **Power-of-a-point reformulation (successful).** Via the median-length formula, $OM = ON \iff \text{pow}(B, \omega) - \text{pow}(C, \omega) = \frac{1}{2}(AB^2 - AC^2)$, where ω is the circumcircle of AKL .
- **Auxiliary points X, Y (successful).** $Y := \text{ray } BK \cap AC$, $X := \text{ray } CL \cap AB$ (crossbar). The three angle hypotheses translate, via directed angles, into three concyclicities: B, X, Y, C on ω_1 ; C, M, K, X on ω_K ; B, N, L, Y on ω_L (with tangency of AB to ω_K at M in the degenerate case $X = M$, and of AC to ω_L at N when $Y = N$).
- **Affine power identity (the key step).** $F(Z) := \text{pow}(Z, \omega) - \text{pow}(Z, \omega_K) - \text{pow}(Z, \omega_L) + \text{pow}(Z, \omega_1)$ is an affine function of Z ; it vanishes at the three non-collinear points A, K, L , hence identically. Evaluating at B and C gives $\text{pow}(B, \omega) = \vec{BM} \cdot \vec{BX}$ and $\text{pow}(C, \omega) =$

$\vec{CN} \cdot \vec{CY}$; together with $AB \cdot AX = AC \cdot AY$ (power of A w.r.t. ω_1) and the median formula this yields $OM = ON$.

- **Dead end avoided:** first tried radical-axis arguments with the second intersection points K^*, L^* of line BY with ω_K and of CX with ω_L ; it works but requires awkward tangency case analysis. The affine function F replaces all of it.
- Numerical verification: `code/verify.py`, `code/check_signs.py`, `code/verify2.py` confirm every claim (several triangles; generic case; both tangency cases $X = M$ and $Y = N$; all signed-angle sign claims; $F \equiv 0$; $OM = ON$) to machine precision.

Current best

Complete rigorous proof below. Core of the argument: with $X = \text{ray } CL \cap AB$, $Y = \text{ray } BK \cap AC$, the hypotheses force B, X, Y, C, C, M, K, X and B, N, L, Y to be concyclic; then the affine function $\text{pow}(\cdot, \omega) - \text{pow}(\cdot, \omega_K) - \text{pow}(\cdot, \omega_L) + \text{pow}(\cdot, \omega_1)$ vanishes at A, K, L , hence everywhere; evaluating at B, C and using $AB \cdot AX = AC \cdot AY$ and the median-length formula gives $OM = ON$.

Full proof

0. Conventions and notation

We identify the plane with $\mathbb{C}$. For $u, v \in \mathbb{C}$ write

$$\langle u, v \rangle = \text{Re}(\bar{u}v), \quad u \times v = \text{Im}(\bar{u}v),$$

the usual dot and cross products of plane vectors.

Unsigned angles. For points $P, Q \neq V$, $\angle PVQ \in [0^\circ, 180^\circ]$ is the ordinary (unsigned) angle, i.e. $\angle PVQ = |\arg \frac{Q-V}{P-V}|$ with $\arg \in (-180^\circ, 180^\circ]$.

Signed ray angles. For nonzero u, v let $\angle(u \rightarrow v) := \arg \frac{v}{u} \in (-180^\circ, 180^\circ]$. For rays from V through P and through Q put $\angle(VP \rightarrow VQ) := \angle((P-V) \rightarrow (Q-V))$. Since $\frac{v}{u} = \frac{\bar{u}v}{|u|^2}$ we have:

(0.1) $|\angle(u \rightarrow v)| = \angle(u, v)$ (the unsigned angle), and if $u \times v \neq 0$ then the sign of $\angle(u \rightarrow v)$ equals the sign of $u \times v$. Moreover $\angle(u \rightarrow w) \equiv \angle(u \rightarrow v) + \angle(v \rightarrow w) \pmod{360^\circ}$ and $\angle(v \rightarrow u) \equiv -\angle(u \rightarrow v) \pmod{360^\circ}$ (because $\arg$ is additive mod 360° under multiplication and $\arg(u/v) = -\arg(v/u) \pmod{360^\circ}$).

Directed line angles. For lines ℓ_1, ℓ_2 with direction vectors $d_1, d_2 \neq 0$ set

$$\angle(\ell_1, \ell_2) := \arg \frac{d_2}{d_1} \pmod{180^\circ}.$$

This is well defined: replacing d_i by a nonzero real multiple changes $\arg \frac{d_2}{d_1}$ by 0° or 180° . It is additive mod 180° , and $\angle(\ell_1, \ell_2) \equiv 0$ iff $\ell_1 \parallel \ell_2$ (including $\ell_1 = \ell_2$). If rays r_1, r_2 span the lines ℓ_1, ℓ_2 , then $\angle(\ell_1, \ell_2) \equiv \angle(r_1 \rightarrow r_2) \pmod{180^\circ}$. For points we write $\angle(PQ, RS)$ for the directed angle from line PQ to line RS .

Power of a point. For a circle Γ with centre Z_0 and radius $\rho > 0$ and a point Z , $\text{pow}(Z, \Gamma) := |Z - Z_0|^2 - \rho^2$.

Sides of a line. For a line through P with direction d , points Z_1, Z_2 are *strictly on the same side* iff $d \times (Z_1 - P)$ and $d \times (Z_2 - P)$ are nonzero of equal sign; *strictly opposite sides* iff nonzero of opposite signs.

Interior of a triangle / of an angle. For an affinely independent triple P, Q, R , the (open) interior of triangle PQR is $\{\alpha P + \beta Q + \gamma R : \alpha, \beta, \gamma > 0, \alpha + \beta + \gamma = 1\}$; “inside triangle PQR ” means lying in this set. For $0^\circ < \angle QVR < 180^\circ$, “ P lies inside angle QVR ” means: P is strictly on the same side of line VQ as R and strictly on the same side of line VR as Q .

Throughout, $b := AC = |C - A|$, $c := AB = |B - A|$, and

$$\theta := \angle KBA = \angle ACL, \quad \varphi := \angle LBK = \angle LNC, \quad \psi := \angle LCK = \angle BMK,$$

ω denotes the circumcircle of triangle AKL and O its centre, $R_\omega = |OA|$ its radius. Since the problem speaks of *triangle* AKL and its circumcentre, the points A, K, L are pairwise distinct and not collinear; this is part of the hypothesis.

Reflections, rotations and translations of the plane preserve all hypotheses and the conclusion, so **we may and do assume that ABC is positively oriented**:

$$\delta := (B - A) \times (C - A) > 0. \quad (0.2)$$

1. Preliminary lemmas

Lemma 1 (line–circle intersections and power). Let Γ be a circle with centre Z_0 , radius $\rho > 0$, and let ℓ be a line. (i) $\ell \cap \Gamma$ has at most two points. (ii) If $M \in \Gamma$, there is exactly one line through M meeting Γ only at M (the **tangent** at M); its direction is $i(M - Z_0)$, i.e. it is the line through M perpendicular to MZ_0 . (iii) If $Z \in \ell$ and ℓ meets Γ at U and V — where in the tangent case we set $U = V$ — then

$$\text{pow}(Z, \Gamma) = \langle U - Z, V - Z \rangle,$$

the product of signed lengths $ZU \cdot ZV$ along ℓ .

Proof. Parametrize $\ell = \{Z + td : t \in \mathbb{R}\}$ with $|d| = 1$, $Z \in \ell$ arbitrary. Then

$$|Z + td - Z_0|^2 - \rho^2 = t^2 + 2t \langle d, Z - Z_0 \rangle + \text{pow}(Z, \Gamma), \quad (1.1)$$

a monic real quadratic in t ; it has at most two roots, proving (i). For (ii): take $Z = M \in \Gamma$; then $\text{pow}(M, \Gamma) = 0$ and the roots are $t = 0$ and $t = -2\langle d, M - Z_0 \rangle$; they coincide iff $\langle d, M - Z_0 \rangle = 0$, i.e. iff $d \perp (M - Z_0)$, i.e. iff d is a real multiple of $i(M - Z_0)$; that determines a unique line through M . For (iii): if $U = Z + t_1 d$, $V = Z + t_2 d$ (with $t_1 = t_2$ in the tangent case, when (1.1) has a double root), then by Vieta $t_1 t_2 = \text{pow}(Z, \Gamma)$, and $\langle U - Z, V - Z \rangle = t_1 t_2 \langle d, d \rangle = t_1 t_2$. $\square$

Lemma 2 (directed inscribed-angle criterion). Let P, Q, R be pairwise distinct, non-collinear points and let Γ be the (unique) circle through them. Then for every point $S \notin \{P, Q\}$:

$$S \in \Gamma \iff \angle(SP, SQ) \equiv \angle(RP, RQ) \pmod{180^\circ}.$$

Proof. Uniqueness of Γ is classical (its centre is the common point of the perpendicular bisectors of PQ and PR , which are not parallel).

Put $w := (Q - R)\overline{(P - R)} \neq 0$. Since $\frac{Q-R}{P-R} = \frac{w}{|P-R|^2}$ and P, Q, R are not collinear, $\arg \frac{Q-R}{P-R} \not\equiv 0 \pmod{180^\circ}$, i.e. $w \notin \mathbb{R}$, so $\text{Im } w \neq 0$.

Define $g : \mathbb{C} \rightarrow \mathbb{R}$ by $g(Z) := \operatorname{Im} [w(P - Z)\overline{(Q - Z)}]$. Expanding $(P - Z)\overline{(Q - Z)} = P\bar{Q} - P\bar{Z} - \bar{Q}Z + |Z|^2$ shows

$$g(Z) = (\operatorname{Im} w)|Z|^2 + \lambda(Z)$$

with λ a real affine function of $Z \in \mathbb{R}^2$. Hence $\frac{1}{\operatorname{Im} w} g(Z) = |Z - Z_1|^2 - \rho_1$ for some $Z_1 \in \mathbb{C}, \rho_1 \in \mathbb{R}$, so the zero set $\{g = 0\}$ is empty, a single point, or a circle. Now

$$g(P) = 0, \quad g(Q) = 0, \quad g(R) = \operatorname{Im} [(Q - R)\overline{(P - R)}(P - R)\overline{(Q - R)}] = \operatorname{Im} [|P - R|^2 |Q - R|^2] = 0.$$

So $\{g = 0\}$ contains three distinct non-collinear points, hence it is a circle through P, Q, R ; by uniqueness $\{g = 0\} = \Gamma$.

Finally, fix $S \notin \{P, Q\}$. All four numbers $Q - S, P - S, Q - R, P - R$ are nonzero, and

$$\angle(SP, SQ) \equiv \angle(RP, RQ) \pmod{180^\circ} \iff \arg \left[\frac{Q-S}{P-S} \right] \equiv \arg \left[\frac{Q-R}{P-R} \right] \pmod{180^\circ} \iff \frac{Q-S}{P-S} \cdot \overline{\left(\frac{Q-R}{P-R} \right)} \in \mathbb{R} \setminus \{0\}.$$

Multiplying by the positive real $|P-S|^2 |P-R|^2$, the last condition is equivalent to $(Q-S)\overline{(P-S)} \cdot \overline{(Q-R)}(P-R) \in \mathbb{R}$, i.e. (taking conjugates) to $w(P-S)\overline{(Q-S)} \in \mathbb{R}$, i.e. to $g(S) = 0$, i.e. to $S \in \Gamma$. $\square$

Lemma 3 (directed tangent-chord). Let Γ be a circle through the pairwise distinct points M, K, C , and let t be the tangent to Γ at M . Then (i) $\angle(t, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$; (ii) if a line ℓ through M satisfies $\angle(\ell, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$, then $\ell = t$; in particular ℓ meets Γ only at M .

Proof. (i) Maps $Z \mapsto aZ + \beta$ ($a \neq 0$) preserve directed line angles (all direction vectors get multiplied by a , so ratios of directions are unchanged), map circles to circles and tangent lines to tangent lines (they are bijections preserving the intersection pattern). Hence we may assume Γ is the unit circle centred at 0. Write $m, k, c \in \Gamma$, so $|m| = |k| = |c| = 1, \bar{m} = 1/m$ etc. By Lemma 1(ii), t has direction im . Consider

$$z := \frac{(k-m)(m-c)}{m(k-c)}.$$

Then

$$\bar{z} = \frac{(\bar{k} - \bar{m})(\bar{m} - \bar{c})}{\bar{m}(\bar{k} - \bar{c})} = \frac{\frac{m-k}{km} \cdot \frac{c-m}{mc}}{\frac{1}{m} \cdot \frac{c-k}{kc}} = \frac{(m-k)(c-m)}{m(c-k)} = -z,$$

so z is purely imaginary; also $z \neq 0$. Hence $z/i \in \mathbb{R} \setminus \{0\}$. Now

$$\angle(t, MK) - \angle(CM, CK) \equiv \arg \frac{k-m}{im} - \arg \frac{k-c}{m-c} = \arg \left[\frac{(k-m)(m-c)}{im(k-c)} \right] = \arg \frac{z}{i} \equiv 0 \pmod{180^\circ}.$$

(ii) From (i), $\angle(\ell, MK) \equiv \angle(t, MK)$, hence $\angle(\ell, t) \equiv 0$, so $\ell \parallel t$; both pass through M , so $\ell = t$. $\square$

Lemma 4 (median-length formula). For any points O, P, Q , if $W = \frac{1}{2}(P + Q)$ then

$$4|OW|^2 = 2|OP|^2 + 2|OQ|^2 - |PQ|^2.$$

Proof. With vectors from O : $4|W - O|^2 = |(P - O) + (Q - O)|^2 = |P - O|^2 + 2\langle P - O, Q - O \rangle + |Q - O|^2$ and $|PQ|^2 = |P - O|^2 - 2\langle P - O, Q - O \rangle + |Q - O|^2$. Add. $\square$

Lemma 5 (angle addition inside an angle). Let V, Q, R be points with $0^\circ < \angle QVR < 180^\circ$ and let P lie inside angle QVR . Then $\angle QVP + \angle PVR = \angle QVR$, and $0^\circ < \angle QVP < \angle QVR$.

Proof. Unsigned angles and sides of lines are invariant under translations, rotations and reflections, so we may place $V = 0$ and assume the unit vector along VQ is $q = 1$ and $\gamma := \angle QVR \in (0^\circ, 180^\circ)$ with unit vector $r = (\cos \gamma, \sin \gamma)$ along VR (reflect in the x -axis if needed). Let $p = (\cos \sigma, \sin \sigma)$, $\sigma \in (-180^\circ, 180^\circ]$, be the unit vector along VP . “ P on the same side of line VQ (the x -axis) as R ” means $\sin \sigma > 0$, so $\sigma \in (0^\circ, 180^\circ)$. “ P on the same side of line VR as Q ” means $r \times p$ and $r \times q$ have the same sign; $r \times q = -\sin \gamma < 0$ and $r \times p = \sin(\sigma - \gamma)$; since $\sigma - \gamma \in (-180^\circ, 180^\circ)$, the condition $\sin(\sigma - \gamma) < 0$ means $\sigma < \gamma$. So $0 < \sigma < \gamma$, and then $\angle QVP = \sigma$, $\angle PVR = \gamma - \sigma$, $\angle QVR = \gamma$. $\square$

2. Position facts

Write points as affine combinations $Z = \alpha_Z A + \beta_Z B + \gamma_Z C$ with $\alpha_Z + \beta_Z + \gamma_Z = 1$ (barycentric coordinates w.r.t. ABC ; not to be confused with the side lengths b, c). From $Z - A = \beta_Z(B - A) + \gamma_Z(C - A)$, $Z - B = \alpha_Z(A - B) + \gamma_Z(C - B)$, $Z - C = \alpha_Z(A - C) + \beta_Z(B - C)$ and the cyclic identity

$$(B - A) \times (C - A) = (C - B) \times (A - B) = (A - C) \times (B - C) = \delta \quad (2.1)$$

(direct expansion of all three gives $A \times B + B \times C + C \times A$), we get

$$(B - A) \times (Z - A) = \gamma_Z \delta, \quad (C - B) \times (Z - B) = \alpha_Z \delta, \quad (A - C) \times (Z - C) = \beta_Z \delta. \quad (2.2)$$

In particular, by (0.2): Z is strictly on the same side of line AB as C iff $\gamma_Z > 0$; strictly on the same side of BC as A iff $\alpha_Z > 0$; strictly on the same side of CA as B iff $\beta_Z > 0$. The interior of triangle ABC is exactly $\{\alpha_Z, \beta_Z, \gamma_Z > 0\}$.

(P1) K lies strictly inside triangle ABC and $K \notin$ lines AB, BC, CM . Indeed $K = \alpha B + \mu M + \gamma C$ with $\alpha, \mu, \gamma > 0$, $\alpha + \mu + \gamma = 1$ (B, M, C are affinely independent since M is an interior point of AB and $C \notin AB$). As $M = \frac{1}{2}(A + B)$,

$$K = \frac{\mu}{2}A + (\alpha + \frac{\mu}{2})B + \gamma C,$$

all coefficients positive, so K is interior to ABC . In coordinates w.r.t. B, M, C : line $BM = AB$ is $\{\gamma = 0\}$, line BC is $\{\mu = 0\}$, line CM is $\{\alpha = 0\}$; the interior has $\alpha, \mu, \gamma > 0$.

(P2) L lies strictly inside triangle ABC and $L \notin$ lines AC, BC, BN . Same argument with $L = \alpha' B + \nu N + \gamma' C$, $N = \frac{1}{2}(A + C)$: $L = \frac{\nu}{2}A + \alpha' B + (\gamma' + \frac{\nu}{2})C$; line $NC = AC$ is $\{\alpha' = 0\}$, line BC is $\{\nu = 0\}$, line BN is $\{\gamma' = 0\}$.

In particular $A \neq K$, $A \neq L$ (A is not interior), $K \notin AB$, $L \notin AC$, so

$$0^\circ < \theta = \angle KBA < 180^\circ, \quad 0^\circ < \varphi = \angle LNC < 180^\circ, \quad 0^\circ < \psi = \angle BMK < 180^\circ. \quad (2.3)$$

(P3) Definition and position of Y and X . The ray from B through K meets segment AC at a point Y with $Y \neq A, C$; the ray from C through L meets segment AB at a point X with $X \neq A, B$.

Proof for Y. Let K have barycentric coordinates $(\alpha_K, \beta_K, \gamma_K)$, all positive, by (P1). First, A and C are strictly on opposite sides of line BK (direction $K - B$):

$$(K-B) \times (A-B) = -(A-B) \times (K-B), \quad (A-B) \times (K-B) = (A-B) \times (K-A) = -(B-A) \times (K-A) = -\gamma_K \delta < 0$$

so $(K-B) \times (A-B) = \gamma_K \delta > 0$; and by (2.2), $(K-B) \times (C-B) = -(C-B) \times (K-B) = -\alpha_K \delta < 0$. (Here $(A-B) \times (K-B) = (A-B) \times (K-A)$ because $(A-B) \times (A-B) = 0$.) Hence segment AC meets line BK in exactly one point Y , and $Y \neq A, C$ (strict signs), $Y \neq B$ (as $B \notin AC$, else $\delta = 0$).

Next, Y lies on the ray from B through K . Write $Y = B + t(K - B)$, $t \neq 0$. On one hand, Y has barycentric coordinates $(1-s, 0, s)$ with $s \in (0, 1)$, so by (2.2) $(B-A) \times (Y-A) = s\delta > 0$. On the other hand $Y-A = (B-A) + t(K-B)$ and $(B-A) \times (K-B) = (B-A) \times (K-A) = \gamma_K \delta$, so $(B-A) \times (Y-A) = t\gamma_K \delta$. Comparing, $t > 0$.

Proof for X. Symmetric. Let L have barycentric coordinates $(\alpha_L, \beta_L, \gamma_L)$, all positive. Then $(L-C) \times (A-C) = -(A-C) \times (L-C) = -\beta_L \delta < 0$ by (2.2), while from $L-C = \alpha_L(A-C) + \beta_L(B-C)$ and (2.1), $(L-C) \times (B-C) = \alpha_L(A-C) \times (B-C) = \alpha_L \delta > 0$. So A, B are strictly on opposite sides of line CL , and segment AB meets line CL in a single point $X \neq A, B$. Writing $X = C + t(L-C)$, and noting that X has barycentric coordinates $(1-s, s, 0)$ with $s \in (0, 1)$, (2.2) gives $(A-C) \times (X-C) = \beta_X \delta = s\delta > 0$, while $(A-C) \times (X-C) = t(A-C) \times (L-C) = t\beta_L \delta$; hence $t > 0$. $\square$

Introduce the signed coordinates along the two sides:

$$x := AX \in (0, c), \quad y := AY \in (0, b),$$

so that on the line AB (coordinatized from A towards B) the points A, X, M, B sit at $0, x, \frac{c}{2}, c$, and on line AC the points A, Y, N, C sit at $0, y, \frac{b}{2}, b$.

Note for later use: since Y lies on ray BK and $Y \neq B$, **line** $BY = \mathbf{line} BK$; since X lies on ray CL and $X \neq C$, **line** $CX = \mathbf{line} CL$. Also $X \neq Y$ (lines AB and AC meet only at A , and $X \neq A$).

3. Translating the hypotheses into directed angles

By (P1)–(P2), K and L are interior to ABC ; let their barycentric coordinates be positive. We use (2.2) and (0.1) repeatedly. Below, “ Z is on the C -side of AB ” abbreviates “strictly on the same side of line AB as C ”, etc.

(S1) If Z is on the C -side of AB (i.e. $\gamma_Z > 0$), then $\sphericalangle(BA \rightarrow BZ) = -\angle ABZ$ and $\sphericalangle(MB \rightarrow MZ) = +\angle BMZ$. *Proof.* $(A-B) \times (Z-B) = -(B-A) \times (Z-B) = -(B-A) \times (Z-A) = -\gamma_Z \delta < 0$, so by (0.1) the signed angle $\sphericalangle(BA \rightarrow BZ)$ is negative, of absolute value $\angle ABZ$. Also $B-M = \frac{1}{2}(B-A)$ and $(B-A) \times (Z-M) = (B-A) \times (Z-A) - (B-A) \times (M-A) = \gamma_Z \delta - 0 > 0$, so $\sphericalangle(MB \rightarrow MZ) = +\angle BMZ$. $\square$

(S2) If Z is on the B -side of AC (i.e. $\beta_Z > 0$), then $\sphericalangle(CA \rightarrow CZ) = +\angle ACZ$ and $\sphericalangle(NC \rightarrow NZ) = -\angle CNZ$. *Proof.* By (2.2), $(A-C) \times (Z-C) = \beta_Z \delta > 0$, so by (0.1) $\sphericalangle(CA \rightarrow CZ) = +\angle ACZ$. For the second claim: $C-N = \frac{1}{2}(C-A)$, and since $N-A$ is parallel to $C-A$,

$$(C-N) \times (Z-N) = \frac{1}{2}(C-A) \times (Z-N) = \frac{1}{2}(C-A) \times (Z-A) = -\frac{1}{2}(A-C) \times (Z-A) = -\frac{1}{2}\beta_Z \delta < 0,$$

where $(A - C) \times (Z - A) = (A - C) \times (Z - C) + (A - C) \times (C - A) = \beta_Z \delta$. So $\sphericalangle(NC \rightarrow NZ) = -\sphericalangle CNZ$. $\square$

Both K and L are on the C -side of AB and on the B -side of AC (interior points; (2.2)). Hence:

$$\sphericalangle(BA \rightarrow BK) = -\sphericalangle ABK = -\theta, \quad \sphericalangle(BA \rightarrow BL) = -\sphericalangle ABL, \quad (3.1)$$

$$\sphericalangle(CA \rightarrow CL) = +\sphericalangle ACL = +\theta, \quad \sphericalangle(CA \rightarrow CK) = +\sphericalangle ACK, \quad (3.2)$$

$$\sphericalangle(MB \rightarrow MK) = +\sphericalangle BMK = +\psi, \quad (3.3)$$

$$\sphericalangle(NC \rightarrow NL) = -\sphericalangle CNL = -\sphericalangle LNC = -\varphi. \quad (3.4)$$

Use of “ K inside angle LBA ”. Since L is interior to ABC , L is inside angle ABC (same side of AB as C and of BC as A , by (2.2)); by Lemma 5 applied to angle ABC ($0^\circ < \angle ABC < 180^\circ$ as A, B, C form a triangle), $0^\circ < \angle LBA < \angle ABC < 180^\circ$. Applying Lemma 5 to angle LBA and the point K inside it:

$$\sphericalangle ABL = \sphericalangle ABK + \sphericalangle KBL = \theta + \varphi', \quad \varphi' := \sphericalangle LBK.$$

By hypothesis $\varphi' = \sphericalangle LNC = \varphi$. Hence, using (0.1) and (3.1),

$$\sphericalangle(BK \rightarrow BL) \equiv \sphericalangle(BA \rightarrow BL) - \sphericalangle(BA \rightarrow BK) = -(\theta + \varphi) + \theta = -\varphi \pmod{360^\circ}. \quad (3.5)$$

Use of “ L inside angle ACK ”. K is interior to ABC , hence inside angle ACB , so $0^\circ < \angle ACK < \angle ACB < 180^\circ$ by Lemma 5. Applying Lemma 5 to angle ACK and the point L inside it: $\angle ACK = \angle ACL + \angle LCK = \theta + \psi'$ with $\psi' := \angle LCK = \angle BMK = \psi$ by hypothesis. Hence by (3.2),

$$\sphericalangle(CL \rightarrow CK) \equiv \sphericalangle(CA \rightarrow CK) - \sphericalangle(CA \rightarrow CL) = (\theta + \psi) - \theta = +\psi \pmod{360^\circ}. \quad (3.6)$$

Two consequences (“nondegeneracy”): since $0 < \varphi < 180^\circ$ and $0 < \psi < 180^\circ$ by (2.3),

$$L \notin \text{line } BK \quad (\text{by (3.5)}), \quad K \notin \text{line } CL \quad (\text{by (3.6)}), \quad (3.7)$$

because if L lay on line BK (note $L \neq B$: L is interior to ABC), the ray BL would coincide with ray BK or its opposite, forcing $\sphericalangle(BK \rightarrow BL) \in \{0^\circ, 180^\circ\}$, contradicting (3.5) since $-\varphi \not\equiv 0^\circ, 180^\circ \pmod{360^\circ}$; likewise for K and line CL (note $K \neq C$), using (3.6).

4. The three circles

Recall line $BY = \text{line } BK$ and line $CX = \text{line } CL$.

Claim 4.1 (ω_1). *The points B, C, X, Y lie on one circle ω_1 .*

Proof. The points X, Y, B are pairwise distinct and non-collinear: if B were on line XY then (as $X \neq B$) line $XY = \text{line } XB = AB \ni Y$, contradicting $Y \notin AB$ (indeed Y lies on segment AC with $Y \neq A$, and $AB \cap AC = \{A\}$ because A, B, C are not collinear). Also $C \notin \{X, Y\}$. Now compute directed line angles mod 180° :

$$\sphericalangle(BX, BY) \equiv \sphericalangle(BA \rightarrow BK) \equiv -\theta \quad (\text{line } BX = AB \text{ since } X \in AB, X \neq B; \text{ line } BY = \text{line } BK),$$

$\angle(CX, CY) \equiv -\angle(CY, CX) \equiv -\angle(CA \rightarrow CL) \equiv -\theta$ (line $CY = AC$ since $Y \in AC$, $Y \neq C$; line $CX =$ line

So $\angle(CX, CY) \equiv \angle(BX, BY) \pmod{180^\circ}$. By Lemma 2 (with $P = X$, $Q = Y$, $R = B$, $S = C$), C lies on the circle ω_1 through X, Y, B . $\square$

Since $X, B \in \omega_1$ are distinct points of line AB , and $Y, C \in \omega_1$ are distinct points of line AC , Lemma 1(i),(iii) give

$$\text{pow}(A, \omega_1) = xc = yb. \quad (4.1)$$

(Signed products: both X and B lie on the ray from A towards B , etc.) Also

$$\text{pow}(B, \omega_1) = 0, \quad \text{pow}(C, \omega_1) = 0, \quad (4.2)$$

and, since $B \neq Y$ are two points of ω_1 on line BK and $C \neq X$ are two points of ω_1 on line CL ,

$$\text{pow}(K, \omega_1) = \langle B - K, Y - K \rangle, \quad \text{pow}(L, \omega_1) = \langle C - L, X - L \rangle. \quad (4.3)$$

Claim 4.2 (ω_K). *Let ω_K be the circumcircle of C, M, K (these are non-collinear: $K \notin$ line CM and $K \notin AB \supseteq \{M\}$ by (P1)). Then line AB meets ω_K exactly in the multiset $\{M, X\}$: if $X \neq M$ then $\omega_K \cap AB = \{M, X\}$; if $X = M$ then AB is tangent to ω_K at M .*

Proof. By (3.3) and (3.6), $\text{mod } 180^\circ$:

$$\angle(CL, CK) \equiv \psi \equiv \angle(MB, MK) = \angle(AB, MK). \quad (4.4)$$

Case $X \neq M$. The points K, X, C are pairwise distinct ($K \notin AB \ni X$; $C \notin AB$; $K \neq C$) and non-collinear: line $XC =$ line CL and $K \notin$ line CL by (3.7). Also $M \notin \{K, X\}$ ($M \in AB$, $K \notin AB$; $X \neq M$ by assumption). Using line $CX =$ line CL and line $MX = AB =$ line MB , (4.4) reads

$$\angle(CK, CX) \equiv -\psi \equiv \angle(MK, MX) \pmod{180^\circ}.$$

By Lemma 2 (with $P = K$, $Q = X$, $R = C$, $S = M$), M lies on the circle through K, X, C ; that circle passes through the non-collinear points C, M, K , hence equals ω_K ; thus $X \in \omega_K$. Since $M \neq X$ are two points of ω_K on AB , Lemma 1(i) gives $\omega_K \cap AB = \{M, X\}$.

Case $X = M$. Then $M \in$ line CL , so (as $C \neq M$) line $CL =$ line CM , and (4.4) becomes $\angle(AB, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$. By Lemma 3(ii) (applied to $\omega_K \ni M, K, C$ and $\ell = AB$), AB is tangent to ω_K at M . $\square$

By Lemma 1(iii) (line AB through A and through B , meeting ω_K at M and X , coincident if $X = M$):

$$\text{pow}(A, \omega_K) = \frac{c}{2}x, \quad \text{pow}(B, \omega_K) = \left(\frac{c}{2} - c\right)(x - c) = \frac{c}{2}(c - x), \quad (4.5)$$

$$\text{pow}(K, \omega_K) = 0, \quad (4.6)$$

and, since $C \neq X$ are two points of ω_K on line CL ,

$$\text{pow}(L, \omega_K) = \langle C - L, X - L \rangle. \quad (4.7)$$

Claim 4.3 (ω_L). *Let ω_L be the circumcircle of B, N, L (non-collinear: $L \notin$ line BN , $L \notin AC \supseteq \{N\}$ by (P2)). Then line AC meets ω_L exactly in the multiset $\{N, Y\}$ (tangency at N if $Y = N$).*

Proof. By (3.4) and (3.5), mod 180° :

$$\angle(BK, BL) \equiv -\varphi \equiv \angle(NC, NL) = \angle(AC, NL). \quad (4.8)$$

Case $Y \neq N$. The points L, Y, B are pairwise distinct and non-collinear (line $BY = \text{line } BK$ and $L \notin \text{line } BK$ by (3.7); $L \notin AC \ni Y$). Also $N \notin \{L, Y, B\}$. Using line $BY = \text{line } BK$ and line $NY = AC = \text{line } NC$, (4.8) reads

$$\angle(BL, BY) \equiv \varphi \equiv \angle(NL, NY) \pmod{180^\circ}.$$

By Lemma 2 (with $P = L$, $Q = Y$, $R = B$, $S = N$), N lies on the circle through L, Y, B , which then equals ω_L ; so $Y \in \omega_L$ and $\omega_L \cap AC = \{N, Y\}$.

Case $Y = N$. Then $N \in \text{line } BK$, so line $BK = \text{line } BN$, and (4.8) becomes $\angle(AC, NL) \equiv \angle(BN, BL) \pmod{180^\circ}$. By Lemma 3(ii) (with the circle $\omega_L \ni N, L, B$ and $\ell = AC$), AC is tangent to ω_L at N . $\square$

Consequently

$$\text{pow}(A, \omega_L) = \frac{b}{2}y, \quad \text{pow}(C, \omega_L) = \frac{b}{2}(b-y), \quad \text{pow}(L, \omega_L) = 0, \quad \text{pow}(B, \omega_L) = 0, \quad (4.9)$$

and, since $B \neq Y$ are two points of ω_L on line BK ,

$$\text{pow}(K, \omega_L) = \langle B - K, Y - K \rangle. \quad (4.10)$$

5. The affine power identity

Define $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(Z) := \text{pow}(Z, \omega) - \text{pow}(Z, \omega_K) - \text{pow}(Z, \omega_L) + \text{pow}(Z, \omega_1).$$

F is affine. For any circle Γ with centre Z_Γ and radius ρ_Γ , $\text{pow}(Z, \Gamma) = |Z|^2 - 2\langle Z_\Gamma, Z \rangle + |Z_\Gamma|^2 - \rho_\Gamma^2$. In F the $|Z|^2$ -terms cancel ($+1 - 1 - 1 + 1 = 0$), so $F(Z) = \langle u, Z \rangle + v$ for a fixed vector u and scalar v .

$F(A) = 0$: by $\text{pow}(A, \omega) = 0$, (4.5), (4.9), (4.1):

$$F(A) = 0 - \frac{c}{2}x - \frac{b}{2}y + cx = \frac{cx}{2} - \frac{by}{2} = 0,$$

using $cx = by$ from (4.1).

$F(K) = 0$: $\text{pow}(K, \omega) = 0$ ($K \in \omega$), $\text{pow}(K, \omega_K) = 0$, and by (4.3), (4.10) $\text{pow}(K, \omega_1) = \text{pow}(K, \omega_L) = \langle B - K, Y - K \rangle$; the last two cancel.

$F(L) = 0$: $\text{pow}(L, \omega) = 0$, $\text{pow}(L, \omega_L) = 0$, and by (4.3), (4.7) $\text{pow}(L, \omega_1) = \text{pow}(L, \omega_K) = \langle C - L, X - L \rangle$.

Hence $F \equiv 0$: A, K, L are not collinear, so $K - A$ and $L - A$ are linearly independent; from $F(A) = F(K) = F(L) = 0$ we get $\langle u, K - A \rangle = \langle u, L - A \rangle = 0$, so $u = 0$, and then $v = F(A) = 0$.

Evaluating F at B and C and using (4.2), (4.5), (4.9):

$$\text{pow}(B, \omega) = \text{pow}(B, \omega_K) + \text{pow}(B, \omega_L) - \text{pow}(B, \omega_1) = \frac{c}{2}(c - x), \quad (5.1)$$

$$\text{pow}(C, \omega) = \text{pow}(C, \omega_K) + \text{pow}(C, \omega_L) - \text{pow}(C, \omega_1) = \frac{b}{2}(b - y). \quad (5.2)$$

6. Conclusion

O is the centre of ω and $R_\omega = |OA|$ its radius, so $|OB|^2 = \text{pow}(B, \omega) + R_\omega^2$ and $|OC|^2 = \text{pow}(C, \omega) + R_\omega^2$. By Lemma 4 applied to the midpoints M of AB and N of AC :

$$4OM^2 = 2OA^2 + 2OB^2 - c^2, \quad 4ON^2 = 2OA^2 + 2OC^2 - b^2.$$

Subtracting and using (5.1), (5.2) and $cx = by$ from (4.1):

$$\begin{aligned} 4(OM^2 - ON^2) &= 2(OB^2 - OC^2) - c^2 + b^2 = 2[\text{pow}(B, \omega) - \text{pow}(C, \omega)] - c^2 + b^2 \\ &= c(c - x) - b(b - y) - c^2 + b^2 = -cx + by = 0. \end{aligned}$$

Therefore $OM = ON$. ■

Remarks on hypothesis usage

- $\angle KBA = \angle ACL \rightarrow$ Claim 4.1; $\angle LCK = \angle BMK$ together with “ L inside angle ACK ” $\rightarrow$ Claim 4.2; $\angle LBK = \angle LNC$ together with “ K inside angle LBA ” $\rightarrow$ Claim 4.3.
- “ K inside triangle BMC ” is used exactly for: K interior to ABC (sign facts (S1)–(S2), crossbar (P3)) and $K \notin$ line CM (ω_K nondegenerate). Similarly for L and line BN .
- That M, N are midpoints enters only through $AM = MB = \frac{c}{2}$, $AN = NC = \frac{b}{2}$ in (4.5), (4.9) and through Lemma 4.
- Numerical verification of every claim (generic and both tangency cases): `code/verify.py`, `code/check_signs.py`, `code/verify2.py`.

IMO 2026 — Problem 3

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Solution document

Status

solved

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Approaches tried

- **Reduction of the picking phase** to the classical alternating-selection game: value to the first player = sum of odd-ranked pieces (sorted descending); equivalently $V = (1 + D)/2$ where D is the alternating sum of the sorted piece lengths. Proved (Lemma 2 below) and verified by minimax search (`code/verify_lemmaA.py`).

- **Brute force on integer grids** (`code/brute.c`): computed max–min D over all Liu Bang configurations and all Xiang Yu refinements for $n = 1$ ($N = 12$), $n = 2$ ($N = 42$), $n = 3$ ($N = 30, 45$), $n = 4$ ($N = 62$), and single-configuration check for $n = 5$ ($N = 126$). All match $D^* = 1/(2^{n+1} - 1)$, i.e. $c = 2^n/(2^{n+1} - 1)$, with the geometric configuration $(2^n, \dots, 2, 1) \cdot u$ among the optimizers.
- **Dead ends (recorded honestly)**: (a) a “parity-measure” integral formula $D = \mu\{t : \#\{\text{parts} > t\} \text{ odd}\}$ — correct but did not directly yield the lower bound; (b) attempts to prove the upper bound by induction on the operation-recursion (halve largest / cut off a copy of the second largest) with the potential $s/(2^m - 1)$ — fails in the “flat top” case without strengthening; (c) an “integrality of the optimal response” reduction — vertices of the arrangement need not be integral. See `approaches/`.
- **Successful ideas**: lower bound by a *tree-pairing* / *signed subset-sum* argument (Lemma 4); upper bound by *subset-sum pigeonhole* + *cross-pairing cuts* (Lemma 5). Both verified numerically (`code/verify_lemmaC.py`, `code/verify_lemmaD.py`, exact rational arithmetic for the constructed strategy).

Current best

Answer:

$$c = \frac{2^n}{2^{n+1} - 1}.$$

Liu Bang’s optimal marking cuts the stick into pieces of lengths $2^n u, 2^{n-1} u, \dots, 2u, u$ where $u = \frac{1}{2^{n+1}-1}$; he then always claims a longest unclaimed piece. Complete proof below.

Full proof

Throughout, fix $n \geq 1$ and set

$$u = \frac{1}{2^{n+1} - 1}, \quad c = \frac{2^n}{2^{n+1} - 1} = \frac{1 + u}{2}.$$

0. Setup and notation

Liu Bang (L) marks a set P_L of at most n points in the open interval $(0, 1)$; then Xiang Yu (Y) marks a set P_Y of at most n further points of $(0, 1)$, disjoint from P_L . The stick is cut at $P_L \cup P_Y$; the resulting pieces are the maximal subintervals, a finite multiset of positive lengths summing to 1. Then the players alternately claim pieces, L first.

For a finite multiset A of nonnegative reals let $a_1 \geq a_2 \geq \dots \geq a_M$ be its elements in non-increasing order (this **sequence of values** is uniquely determined by A). Define

$$O(A) = \sum_{i \text{ odd}} a_i, \quad E(A) = \sum_{i \text{ even}} a_i, \quad D(A) = \sum_{i=1}^M (-1)^{i-1} a_i = O(A) - E(A).$$

Since $O(A) + E(A) = \Sigma(A)$ (the total), if $\Sigma(A) = 1$ then

$$O(A) = \frac{1 + D(A)}{2}. \quad (0.1)$$

Grouping consecutive terms gives

$$D(A) = \sum_{i=1}^{\lfloor M/2 \rfloor} (a_{2i-1} - a_{2i}) + \begin{cases} a_M, & M \text{ odd} \\ 0, & M \text{ even,} \end{cases} \quad (0.2)$$

so $0 \leq D(A)$, and trivially $D(A) \leq \Sigma(A)$.

1. Lemma 1 (equal pairs cancel)

For any finite multiset X and any $y \geq 0$: $D(X \uplus \{y, y\}) = D(X)$.

Proof. Let $x_1 \geq \dots \geq x_r$ be the sorted values of X and let p be the number of $x_i \geq y$. Then the sorted value sequence of $X \uplus \{y, y\}$ is $x_1, \dots, x_p, y, y, x_{p+1}, \dots, x_r$. Its alternating sum is

$$\sum_{i \leq p} (-1)^{i-1} x_i + (-1)^p y + (-1)^{p+1} y + \sum_{i > p} (-1)^{(i+2)-1} x_i = D(X). \quad \blacksquare$$

By induction, adjoining any number of equal pairs to a multiset does not change D ; in particular a multiset consisting only of equal pairs and a sub-multiset L has $D = D(L) \leq \Sigma(L)$.

2. Lemma 2 (value of the picking phase)

Let A be the multiset of piece lengths, $\Sigma(A) = 1$. In the alternating claiming phase: 1. The first player has a strategy guaranteeing himself a total $\geq O(A)$, against every play of the opponent. 2. The second player has a strategy guaranteeing herself a total $\geq E(A)$, against every play of the opponent.

Consequently, under (0.1), the first player can guarantee exactly $\frac{1+D(A)}{2}$ from the picking phase and no more.

Proof. Write $A : a_1 \geq \dots \geq a_M$. We first record two inequalities. For $2 \leq j \leq M$, removing a_1 and a_j from A leaves the sorted sequence $a_2, \dots, a_{j-1}, a_{j+1}, \dots, a_M$, in which a_i has rank $i - 1$ for $i < j$ and rank $i - 2$ for $i > j$; hence

$$O(A \setminus \{a_1, a_j\}) = \sum_{\substack{2 \leq i \leq j-1 \\ i \text{ even}}} a_i + \sum_{\substack{i > j \\ i \text{ odd}}} a_i.$$

Comparing with $O(A) - a_1 = \sum_{3 \leq i \leq j, i \text{ odd}} a_i + \sum_{i > j, i \text{ odd}} a_i$ and pairing terms,

$$O(A \setminus \{a_1, a_j\}) - (O(A) - a_1) = \begin{cases} (a_2 - a_3) + \dots + (a_{j-2} - a_{j-1}), & j \text{ even} \\ (a_2 - a_3) + \dots + (a_{j-1} - a_j), & j \text{ odd} \end{cases} \geq 0. \quad (2.1)$$

Similarly, removing one element a_j leaves a_i at rank i for $i < j$ and rank $i - 1$ for $i > j$, so $O(A \setminus \{a_j\}) = \sum_{i < j, i \text{ odd}} a_i + \sum_{i > j, i \text{ even}} a_i$, and

$$O(A \setminus \{a_j\}) - E(A) = \begin{cases} (a_1 - a_2) + \dots + (a_{j-2} - a_{j-1}), & j \text{ odd} \\ (a_1 - a_2) + \dots + (a_{j-1} - a_j), & j \text{ even} \end{cases} \geq 0. \quad (2.2)$$

Proof of 1. Strategy: always claim a currently longest unclaimed piece. We show by strong induction on $|B|$: if it is your turn and the unclaimed multiset is B , this strategy nets you at least $O(B)$ from B . For $|B| \leq 1$ this is clear. Otherwise you take b_1 (a largest element). If nothing remains, you got $b_1 = O(B)$. Otherwise the opponent takes some b_j ($j \geq 2$), and by induction you gain at least $O(B \setminus \{b_1, b_j\}) \geq O(B) - b_1$ by (2.1) from the rest. Total $\geq O(B)$.

Proof of 2. Strategy: after the opponent's first claim a_j , it is your turn on $A \setminus \{a_j\}$; play the greedy strategy of part 1 to get at least $O(A \setminus \{a_j\}) \geq E(A)$ by (2.2).

Finally, part 1 gives the first player $\geq O(A)$; part 2 gives the second player $\geq E(A)$, i.e. caps the first player at $1 - E(A) = O(A)$. ■

Thus, once the cutting is done and produces the multiset A , the quantity Liu Bang can guarantee from that position is exactly $\frac{1+D(A)}{2}$, and Xiang Yu can prevent more. So the whole game reduces to:

Reduced game. L chooses a multiset $q_1, \dots, q_m$ ($m \leq n+1$) of positive reals summing to 1 (the pieces cut by his marks). Y then refines it: she distributes $K \leq n$ cuts among the pieces, splitting a piece of length q receiving k cuts into $k+1$ positive parts summing to q . If A is the final multiset of parts, L's guaranteed amount is $\frac{1+D(A)}{2}$: he wants $D(A)$ large, she wants it small.

(Y's marks are automatically distinct from L's, because each of her marks is interior to one of L's pieces; conversely any refinement of the pieces is realizable by valid marks.)

3. The signed subset-sum property of the geometric configuration

Let L's configuration be the **geometric configuration**

$$q_j = 2^j u \quad (j = 0, 1, \dots, n), \quad \sum_j q_j = (2^{n+1} - 1)u = 1.$$

Its key property: for every nonempty $J \subseteq \{0, \dots, n\}$ and signs $s_j \in \{\pm 1\}$,

$$\left| \sum_{j \in J} s_j 2^j u \right| \geq u. \quad (3.1)$$

Indeed, with $j^* = \max J$, $|\sum_{j \in J} s_j 2^j| \geq 2^{j^*} - \sum_{j < j^*} 2^j = 1$.

4. Lemma 4 (lower bound: Y cannot push D below u)

Let A be any refinement of the geometric configuration obtained by at most n cuts. Then $D(A) \geq u$.

Proof. Say Y used $K \leq n$ cuts, so A has $M = (n+1) + K \leq 2n+1$ parts; each part lies inside a unique original piece. Fix a non-increasing enumeration $a_1 \geq a_2 \geq \dots \geq a_M$ of the **parts as physical objects** (ties broken arbitrarily); $D(A)$ is its alternating sum. Form the pairs $P_i = \{a_{2i-1}, a_{2i}\}$ for $1 \leq i \leq \lfloor M/2 \rfloor$, with slack $\varepsilon_i = a_{2i-1} - a_{2i} \geq 0$; if M is odd, a_M is unpaired. By (0.2),

$$D(A) = \sum_i \varepsilon_i + [M \text{ odd}] a_M. \quad (4.1)$$

Build a multigraph G with vertex set $V = \{0, 1, \dots, n\}$ (the original pieces) and one edge per pair P_i , joining the two pieces containing the two parts of P_i (a loop if they lie in the same piece). Then

$$|E| = \left\lfloor \frac{M}{2} \right\rfloor \leq n < n+1 = |V|.$$

Every connected multigraph on v vertices has at least $v-1$ edges, with equality only if it is a **simple tree** (take a spanning tree, which uses $v-1$ loop-free, pairwise distinct edges; with only

$v - 1$ edges in total there is nothing else). If every component of G contained a cycle or loop it would have at least as many edges as vertices, forcing $|E| \geq |V|$ — impossible. Hence some component $T = (V_T, E_T)$ is a simple tree (possibly a single vertex).

Being a tree, T admits a proper 2-colouring $s : V_T \rightarrow \{\pm 1\}$ (endpoints of every edge get opposite signs). Consider

$$\Xi := \sum_{j \in V_T} s(j) q_j = \sum_{j \in V_T} s(j) 2^j u.$$

By (3.1), $|\Xi| \geq u$.

Now expand Ξ over parts: q_j is the sum of the parts inside piece j . Consider any part x inside a piece of V_T . Either x is the unpaired a_M , or x belongs to a pair P_i ; the edge P_i is incident to V_T , hence (as T is a connected component) both endpoints of P_i lie in V_T , and they are distinct pieces because T has no loops. Consequently

$$\Xi = \sum_{P_i \in E_T} \left(s(\pi(a_{2i-1})) a_{2i-1} + s(\pi(a_{2i})) a_{2i} \right) + [M \text{ odd}, \pi(a_M) \in V_T] s(\pi(a_M)) a_M,$$

where $\pi(x)$ denotes the piece containing x . For each edge $P_i \in E_T$ the two signs are opposite, so the corresponding term has absolute value $|a_{2i-1} - a_{2i}| = \varepsilon_i$. Hence, using (4.1),

$$u \leq |\Xi| \leq \sum_{P_i \in E_T} \varepsilon_i + [M \text{ odd}] a_M \leq D(A). \quad \blacksquare$$

Lower bound conclusion. Liu Bang marks the n interior points $2^n u, (2^n + 2^{n-1})u, \dots, (2^{n+1} - 2)u$, cutting the stick into the geometric pieces $2^n u, \dots, 2u, u$. Whatever at most n points Xiang Yu adds, the final multiset A satisfies $D(A) \geq u$ by Lemma 4, and by Lemma 2(1) greedy claiming guarantees Liu Bang at least $\frac{1+D(A)}{2} \geq \frac{1+u}{2} = c$, **against every behaviour of Xiang Yu**.

5. Lemma 5 (upper bound: Y can always force $D \leq u$)

Whatever at most n points L marks, Y can mark at most n points so that the final multiset A satisfies $D(A) \leq u$.

Proof. Let L 's marks create pieces $q_1, \dots, q_m > 0$, $m \leq n + 1$, $\sum q_i = 1$.

Case $m \leq n$. Y marks the midpoint of every piece ($m \leq n$ interior points, pairwise distinct). The final multiset consists of the equal pairs $\{q_i/2, q_i/2\}$, so $D(A) = D(\emptyset) = 0$ by Lemma 1.

Case $m = n + 1$. Consider the 2^m subset sums $\sigma(S) = \sum_{i \in S} q_i$, $S \subseteq \{1, \dots, m\}$. They lie in $[0, 1]$, with $\sigma(\emptyset) = 0$ and $\sigma(\{1, \dots, m\}) = 1$; sorting them, the $2^m - 1$ consecutive gaps sum to 1, so some gap is at most $\frac{1}{2^m - 1} = u$: there are $S \neq S'$ with $0 \leq \sigma(S') - \sigma(S) \leq u$. Put $A^* = S' \setminus S$ and $B^* = S \setminus S'$; these are disjoint, not both empty, and

$$\Delta := \sigma(A^*) - \sigma(B^*) = \sigma(S') - \sigma(S) \in [0, u].$$

Since $\Delta \geq 0$, $A^* = \emptyset$ would force $B^* = \emptyset$; hence $A^* \neq \emptyset$.

Sub-case $B^ = \emptyset$* (so $\sigma(A^*) = \Delta \leq u$). Y halves each of the $m - |A^*| \leq m - 1 = n$ pieces not in A^* and marks nothing else. The final multiset is (equal pairs) $\uplus \{q_i : i \in A^*\}$, so by Lemma 1, $D(A) = D(\{q_i\}_{i \in A^*}) \leq \sigma(A^*) = \Delta \leq u$.

Sub-case $A^*, B^* \neq \emptyset$. Write $A^* = \{i_1, \dots, i_p\}$, $B^* = \{j_1, \dots, j_q\}$, and set partial sums

$$\alpha_0 = 0, \alpha_k = q_{i_1} + \dots + q_{i_k} \quad (1 \leq k \leq p), \quad \beta_0 = 0, \beta_k = q_{j_1} + \dots + q_{j_k} \quad (1 \leq k \leq q),$$

so $\alpha_p = \sigma(A^*)$ and $\beta_q = \sigma(B^*) = \alpha_p - \Delta$. Identify the pieces of A^* , concatenated, with the interval $[0, \alpha_p]$ (piece $i_k \leftrightarrow [\alpha_{k-1}, \alpha_k]$), and the pieces of B^* with $[0, \beta_q]$. Y marks: - the midpoint of each of the $m - p - q$ pieces outside $A^* \cup B^*$; - inside the A^* -row: every point of $\{\beta_1, \dots, \beta_{q-1}, \beta_q\} \cap (0, \alpha_p)$ that is **not** equal to some α_k — at most q points, each interior to exactly one A^* -piece; - inside the B^* -row: every point of $\{\alpha_1, \dots, \alpha_{p-1}\} \cap (0, \beta_q)$ that is not equal to some β_k — at most $p - 1$ points.

Total marks $\leq (m - p - q) + q + (p - 1) = m - 1 = n$. All are interior points of L's pieces (so distinct from L's marks), and pairwise distinct (distinct positions along the rows; positions in different pieces are different points of the stick).

Let $\Gamma : 0 = \gamma_0 < \gamma_1 < \dots < \gamma_r = \beta_q$ be the sorted distinct values of $(\{\alpha_k\}_{k=0}^p \cup \{\beta_k\}_{k=0}^q) \cap [0, \beta_q]$. By construction, the cut grid of the A^* -row restricted to $[0, \beta_q]$ is exactly Γ (all $\beta_k \leq \beta_q$ are either marked in the A^* -row or coincide with an α_k ; note $\beta_q \leq \alpha_p$), and the cut grid of the B^* -row on $[0, \beta_q]$ is also exactly Γ . Hence: - for each $0 \leq t < r$, the interval (γ_t, γ_{t+1}) produces **one part of length $\gamma_{t+1} - \gamma_t$ on the A^* -side and one on the B^* -side** — an equal pair; - no A^* -part straddles β_q (the point β_q belongs to the A^* -grid, or equals α_p); the remaining A^* -parts fill $[\beta_q, \alpha_p]$, a leftover multiset L with $\Sigma(L) = \Delta$ (empty if $\Delta = 0$); - pieces outside $A^* \cup B^*$ produce equal pairs $\{q_i/2, q_i/2\}$.

The final multiset is therefore (equal pairs) $\uplus L$, and by Lemma 1,

$$D(A) = D(L) \leq \Sigma(L) = \Delta \leq u. \quad \blacksquare$$

Upper bound conclusion. Whatever Liu Bang's marks, Xiang Yu marks as in Lemma 5 and then, in the claiming phase, plays the greedy second-player strategy of Lemma 2(2). She thereby secures at least $E(A) = \frac{1-D(A)}{2} \geq \frac{1-u}{2}$, so Liu Bang ends with at most $\frac{1+D(A)}{2} \leq \frac{1+u}{2} = c$, **no matter how he picks**. (In the case $m \leq n$ he even gets at most $1/2 < c$.)

6. Conclusion

Liu Bang has a strategy (geometric marking + greedy claiming) guaranteeing at least c ; Xiang Yu has, against every strategy of Liu Bang, a reply (Lemma 5 marking + greedy claiming) that caps him at c . Therefore the largest total length Liu Bang can guarantee is exactly

$$c = \frac{2^n}{2^{n+1} - 1}.$$

(Numerically verified: $n = 1 : 2/3$, $n = 2 : 4/7$, $n = 3 : 8/15$, $n = 4 : 16/31$ by exhaustive grid search; $n = 5 : 32/63$ for the geometric configuration; plus 40000-trial Monte Carlo for Lemma 4 and 4000 exact-arithmetic random tests of the Lemma 5 construction.)

IMO 2026 — Problem 4

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Problem

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Approaches tried

- **Cut parametrization** (`lemmas/cut-parametrization.md`): a cut through the vertex with angle A , with $\alpha = \angle BAP \in (0, A)$, produces children with angle multisets $\{B, \alpha, 180^\circ - B - \alpha\}$

and $\{C, A - \alpha, B + \alpha\}$. Verified numerically (`code/verify.py`, `geo_check`). This is the foundation of everything below.

- **Immediate-win analysis:** with generic angles, a single cut can put a multiple of θ into *both* children only if the split angle is a multiple of θ , or if the two supplementary angles at P are both multiples of θ — the latter forces $180^\circ \in \theta\mathbb{N}$. This suggested the dichotomy “ θ divides 180° or not”.
- **Mulan’s strategy for $\theta = 180^\circ/n$** (`approaches/mulan-strategy-divisor-theta.md`): chain lemma (angle $k\theta$ win in $\leq k - 1$ cuts) plus a double-threat cut through the largest angle making both angles at P multiples of θ . Complete; verified exactly for $n = 2, \dots, 8$ against random play.
- **Shan-Yu’s invariant for $180^\circ/\theta \notin \mathbb{Z}$** (`approaches/shanyu-invariant.md`): keep all three angles outside $\theta\mathbb{N}$; a 4-case algebraic check shows this is always possible. Complete; verified exactly for several such θ with adversarial cut values.
- **Dead ends:** order-type invariants for Shan-Yu (“all angles $< \theta$ ”, “all angles $> \theta$ ”, “all in $(\theta, 2\theta)$ ”) — only the first works and only for $\theta > 90^\circ$; all subsumed by the arithmetic invariant.

Current best

Answer: Mulan can guarantee victory exactly for the angles θ such that $180^\circ/\theta$ is an integer, i.e. $\theta = 180^\circ/n$ for some integer $n \geq 2$ (namely $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, \dots\}$). If $n\theta = 180^\circ$, Mulan wins within at most $n - 1$ cuts via a “double threat + chain” strategy; if 180° is not an integer multiple of θ , Shan-Yu survives forever by keeping all three angles outside $\{\theta, 2\theta, 3\theta, \dots\}$, which a short case analysis shows is always possible. Full proof below; all key computations sanity-checked exactly in `code/verify.py`.

Full proof

Throughout, we identify a triangle with the (unordered) triple of its angles, all measured in degrees; the triple consists of positive reals summing to 180° . Let

$$\theta\mathbb{N} := \{k\theta : k = 1, 2, 3, \dots\}$$

denote the set of positive integer multiples of θ . Note that only the finitely many elements of $\theta\mathbb{N} \cap (0^\circ, 180^\circ)$ can occur as angles of a triangle.

Answer. Mulan can guarantee victory in finitely many steps if and only if

$$\theta = \frac{180^\circ}{n} \quad \text{for some integer } n \geq 2,$$

equivalently, iff $180^\circ \in \theta\mathbb{N}$. (Since $0^\circ < \theta < 180^\circ$, the integer $n = 180^\circ/\theta$ automatically satisfies $n \geq 2$.)

Lemma 0 (Cut parametrization)

Let $\mathcal{T}$ be a triangle with angles A, B, C at vertices also named A, B, C . Mulan’s legal cuts through vertex A (i.e. with P in the interior of side BC) are parametrized bijectively by $\alpha := \angle BAP \in (0^\circ, A)$, and the resulting two triangles have angle multisets

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\}, \quad T_2 = \{C, A - \alpha, B + \alpha\}.$$

Proof. As P traverses the open segment BC , the ray AP sweeps the open angular sector at A between the rays AB and AC ; the map $P \mapsto \alpha = \angle BAP$ is a continuous strictly monotone bijection from the interior of BC onto $(0^\circ, A)$. Triangle ABP has angle B at B and α at A , hence angle $180^\circ - B - \alpha$ at P . Triangle APC has angle C at C and $A - \alpha$ at A , hence at P the angle

$$180^\circ - C - (A - \alpha) = (180^\circ - A - B - C) + B + \alpha = B + \alpha. \quad \blacksquare$$

Note for later use: the two angles at P , namely $180^\circ - B - \alpha$ (in T_1) and $B + \alpha$ (in T_2), are supplementary.

A **round** consists of: the stopping check, then (if the game did not stop) one cut by Mulan and one discard by Shan-Yu. “Mulan wins within m cuts” means: whatever Shan-Yu does, the stopping check succeeds after at most m cuts have been made.

Part 1: If $n\theta = 180^\circ$ for an integer $n \geq 2$, Mulan wins (within at most $n - 1$ cuts)

Here $\theta\mathbb{N} \cap (0^\circ, 180^\circ) = \{k\theta : 1 \leq k \leq n - 1\}$.

Lemma 1 (Chain lemma). *For any $\theta \in (0^\circ, 180^\circ)$: if at the start of a round the triangle has an angle equal to $k\theta$ for some integer $k \geq 1$, then Mulan can win within at most $k - 1$ further cuts.*

Proof. Induction on k .

Base $k = 1$: the triangle has an angle exactly θ , so the stopping check ends the game at once; Mulan has won with 0 further cuts.

Step $k \geq 2$: let $A = k\theta$ be such an angle (note $k\theta < 180^\circ$), and let B, C be the other two angles. Since $0^\circ < \theta < k\theta = A$, Lemma 0 allows Mulan to cut through vertex A with $\alpha = \theta$. The children are

$$T_1 = \{B, \theta, 180^\circ - B - \theta\}, \quad T_2 = \{C, (k - 1)\theta, B + \theta\}.$$

If Shan-Yu keeps T_1 , then the next round starts with a triangle having an angle exactly θ , and the game stops with Mulan’s victory: $1 \leq k - 1$ cut was used. If Shan-Yu keeps T_2 , it has the angle $(k - 1)\theta$, so by the induction hypothesis Mulan wins within $k - 2$ further cuts, for a total of at most $k - 1$. $\blacksquare$

Lemma 2 (Double threat). *Assume $n\theta = 180^\circ$, $n \geq 2$. If at the start of a round the triangle has no angle in $\theta\mathbb{N}$, then Mulan has a cut such that **both** children contain an angle of the form $k\theta$ with $1 \leq k \leq n - 1$.*

Proof. Let A be a largest angle of the triangle and B, C the other two, so $A \geq 60^\circ$. We claim there is an integer $k \geq 1$ with

$$C < k\theta < 180^\circ - B. \quad (*)$$

The interval $(C, 180^\circ - B)$ is open with length $180^\circ - B - C = A$.

Case $n \geq 3$: then $\theta \leq 60^\circ \leq A$, and $A \notin \theta\mathbb{N}$ (hypothesis) gives $A \neq \theta$, hence $A > \theta$. Let k be the least positive integer with $k\theta > C$ (it exists since $\theta > 0$). Since $C \notin \theta\mathbb{N}$ and $(k - 1)\theta \leq C$, we get $k\theta \leq C + \theta < C + A = 180^\circ - B$. Thus $(*)$ holds.

Case $n = 2$ (so $\theta = 90^\circ$): we show $B, C < 90^\circ$, whence $(*)$ holds with $k = 1$: indeed $C < 90^\circ$ and $180^\circ - B > 90^\circ$. To see $B, C < 90^\circ$: if $A < 90^\circ$ then all three angles are $< 90^\circ$; $A = 90^\circ$ is impossible because no angle lies in $\theta\mathbb{N} \ni 90^\circ$; and if $A > 90^\circ$ then $B + C = 180^\circ - A < 90^\circ$, so $B, C < 90^\circ$.

Fix such k , and note $k\theta < 180^\circ - B < 180^\circ = n\theta$ forces $1 \leq k \leq n-1$. Mulan cuts through vertex A with

$$\alpha := 180^\circ - B - k\theta.$$

By (*), $0 < \alpha < 180^\circ - B - C = A$, so the cut is legal by Lemma 0. The children are

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\} = \{B, 180^\circ - B - k\theta, k\theta\},$$

$$T_2 = \{C, A - \alpha, B + \alpha\} = \{C, k\theta - C, 180^\circ - k\theta\} = \{C, k\theta - C, (n-k)\theta\},$$

where we used $A - \alpha = A - 180^\circ + B + k\theta = k\theta - C$ and $180^\circ - k\theta = (n-k)\theta$ with $1 \leq n-k \leq n-1$. Thus T_1 contains $k\theta$ and T_2 contains $(n-k)\theta$, both in $\{\theta, \dots, (n-1)\theta\}$. ■

Mulan's complete strategy for $\theta = 180^\circ/n$. At the start of the game (round 1), either the triangle already has an angle $k\theta$ for some $1 \leq k \leq n-1$ — then by Lemma 1 Mulan wins within $k-1 \leq n-2$ cuts — or it has no angle in $\theta\mathbb{N}$. In the latter case Mulan plays the cut of Lemma 2; whichever child Shan-Yu keeps, the next round starts with a triangle containing an angle $k'\theta$ with $1 \leq k' \leq n-1$, and by Lemma 1 Mulan wins within $k'-1 \leq n-2$ further cuts. In total Mulan wins within at most $n-1$ cuts, no matter how Shan-Yu chooses the initial triangle and his discards. This proves Part 1.

Part 2: If $180^\circ \notin \theta\mathbb{N}$, Shan-Yu can avoid defeat forever

Call a triangle **safe** if none of its three angles belongs to $\theta\mathbb{N}$. Since $\theta \in \theta\mathbb{N}$, a safe triangle never triggers the stopping check.

Lemma 3 (Existence of a safe start). *A safe triangle exists.*

Proof. The set $F := \theta\mathbb{N} \cap (0^\circ, 180^\circ)$ is finite. Pick any $A \in (0^\circ, 180^\circ) \setminus F$ (possible: F is finite, the interval is infinite). Now pick $B \in (0^\circ, 180^\circ - A)$ with $B \notin F$ and $180^\circ - A - B \notin F$: this excludes only finitely many values of B from an interval of positive length, so such B exists. Set $C = 180^\circ - A - B > 0$. Then (A, B, C) is safe. ■

Lemma 4 (Closure). *Assume $180^\circ \notin \theta\mathbb{N}$. If the current triangle is safe, then for every legal cut of Mulan, at least one of the two children is safe.*

Proof. Suppose Mulan cuts through the vertex whose angle is A ; let B, C be the other two angles and $\alpha \in (0^\circ, A)$ the cut parameter. By Lemma 0 the children are

$$T_1 = \{B, \alpha, 180^\circ - B - \alpha\}, \quad T_2 = \{C, A - \alpha, B + \alpha\}.$$

By safety, $B \notin \theta\mathbb{N}$ and $C \notin \theta\mathbb{N}$. Hence: - if T_1 is unsafe, then $\alpha \in \theta\mathbb{N}$ or $180^\circ - B - \alpha \in \theta\mathbb{N}$; - if T_2 is unsafe, then $A - \alpha \in \theta\mathbb{N}$ or $B + \alpha \in \theta\mathbb{N}$.

Assume for contradiction both children are unsafe. There are four cases, with $k, m \geq 1$ integers:

1. $\alpha = k\theta$ and $A - \alpha = m\theta$. Then $A = (k+m)\theta \in \theta\mathbb{N}$, contradicting safety of the current triangle.
2. $\alpha = k\theta$ and $B + \alpha = m\theta$. Then $B = (m-k)\theta$; since $B > 0$, $m-k \geq 1$, so $B \in \theta\mathbb{N}$ — contradiction.
3. $180^\circ - B - \alpha = k\theta$ and $A - \alpha = m\theta$. Subtracting: $180^\circ - A - B = (k-m)\theta$, i.e. $C = (k-m)\theta$; since $C > 0$, $k-m \geq 1$, so $C \in \theta\mathbb{N}$ — contradiction.
4. $180^\circ - B - \alpha = k\theta$ and $B + \alpha = m\theta$. Adding: $180^\circ = (k+m)\theta \in \theta\mathbb{N}$ — contradicting the hypothesis of Part 2.

All cases are impossible, so at least one child is safe. ■

Shan-Yu's complete strategy for $180^\circ \notin \theta\mathbb{N}$. He creates a safe initial triangle (Lemma 3). Whenever Mulan cuts, he keeps a safe child, which exists by Lemma 4. By induction, at the start of every round the triangle is safe, hence has no angle equal to θ , hence the stopping check never succeeds. Therefore no strategy of Mulan wins in finitely many steps: against Shan-Yu's strategy the game runs forever. This proves Part 2.

Conclusion

Combining Parts 1 and 2: Mulan can guarantee victory in finitely many steps **iff** 180° is an integer multiple of θ , i.e.

$$\theta = \frac{180^\circ}{n} \text{ for some integer } n \geq 2$$

(equivalently $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, 180^\circ/7, \dots\}$), and when she can, she needs at most $n - 1$ cuts. ■

IMO 2026 — Problem 5

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- **Squared reformulation + substitution** $x = f(y)$ (successful): squaring both inequalities gives $2x^2 + 2f(y)^2 \geq (f(x) + y)^2 \geq 4xf(y)$. Plugging $x = f(y)$ makes the two bounds collide and forces the exact identity $f(f(y)) = 2f(y) - y$. Orbits of f are then arithmetic progressions; positivity forces $f(y) \geq y$.
- **Testing candidates:** $f(x) = kx$ works only for $k = 1$; but $f(x) = x + c$ works for every constant $c \geq 0$ (QM-AM-GM applied to the pair $x, y + c$). Numerically verified in `code/verify.py`; random perturbations of $x + c$ violate one of the two inequalities.
- **Orbit rigidity / asymptotic sandwich** (successful): a point a with gap $d = f(a) - a > 0$ generates the orbit $a + nd$ on which $f(t) = t + d$. Using orbit points as y in both squared inequalities and choosing the orbit point nearest to x gives the two-sided bound $d - \frac{d^2}{4x} \leq$

$f(x) - x \leq d + \frac{d^2}{16x-4d}$ for $x \geq a + d$. Hence all positive gaps are equal and all large enough x have gap exactly d .

- **Downward bootstrap** (successful): if $f(t) = t + d$ for all $t \geq T$, then taking $x = y + d$ in both squared inequalities pins $f(y) = y + d$ for all $y \geq T - d$; induction covers all of $\mathbb{R}_{>0}$.

Current best

Answer: the solutions are exactly the functions $f(x) = x + c$ for a constant $c \geq 0$. Complete proof below: sufficiency is QM–AM–GM applied to the numbers x and $y + c$; necessity follows from the forced identity $f(f(y)) = 2f(y) - y$ (equality at $x = f(y)$), which makes orbits arithmetic progressions, whence $f(y) \geq y$, then an asymptotic sandwich shows the gap $g = f - \text{id}$ is a constant d on $[a + d, \infty)$ for any a with $g(a) = d > 0$, and a downward bootstrap propagates $f = \text{id} + d$ to all of $\mathbb{R}_{>0}$; if $g \equiv 0$ then $f = \text{id}$ (the case $c = 0$).

Full proof

Answer. The functions satisfying the condition are exactly

$$f(x) = x + c \quad (x > 0),$$

where $c \geq 0$ is an arbitrary nonnegative real constant.

Throughout, since all quantities appearing under square roots and on both sides are positive, the given condition is equivalent to the pair of squared inequalities

$$(1) \quad 2x^2 + 2f(y)^2 \geq (f(x) + y)^2, \quad (2) \quad (f(x) + y)^2 \geq 4xf(y),$$

required for all $x, y > 0$.

Part I: every $f(x) = x + c$ with $c \geq 0$ is a solution

Fix $c \geq 0$ and let $f(x) = x + c$; note f maps $\mathbb{R}_{>0}$ to $\mathbb{R}_{>0}$. For $x, y > 0$ put $u = x > 0$ and $v = y + c > 0$. Then

$$\frac{x^2 + f(y)^2}{2} = \frac{u^2 + v^2}{2}, \quad \frac{f(x) + y}{2} = \frac{x + c + y}{2} = \frac{u + v}{2}, \quad xf(y) = uv,$$

so the required chain is exactly the QM–AM–GM inequality for the two positive numbers u, v :

$$\sqrt{\frac{u^2 + v^2}{2}} \geq \frac{u + v}{2} \geq \sqrt{uv},$$

which holds because $2(u^2 + v^2) - (u + v)^2 = (u - v)^2 \geq 0$ and $(u + v)^2 - 4uv = (u - v)^2 \geq 0$. ■

Part II: every solution is of this form

Let f satisfy (1) and (2) for all $x, y > 0$.

Step 1: $f(f(y)) = 2f(y) - y$ for all $y > 0$. Substitute $x = f(y)$ (a legitimate positive value) into (1):

$$2f(y)^2 + 2f(y)^2 \geq (f(f(y)) + y)^2 \implies 2f(y) \geq f(f(y)) + y,$$

using that both sides are positive. Substitute $x = f(y)$ into (2):

$$(f(f(y)) + y)^2 \geq 4f(y) \cdot f(y) \implies f(f(y)) + y \geq 2f(y).$$

Combining, $f(f(y)) + y = 2f(y)$, i.e.

$$(3) \quad f(f(y)) = 2f(y) - y \quad \text{for all } y > 0.$$

Step 2: $f(y) \geq y$; **orbits are arithmetic progressions.** Fix $y > 0$ and define $y_0 = y$, $y_{n+1} = f(y_n)$ ($n \geq 0$); all $y_n > 0$. Applying (3) at the point y_n gives $y_{n+2} = 2y_{n+1} - y_n$, so (y_n) is an arithmetic progression:

$$y_n = y + n g(y), \quad \text{where } g(y) := f(y) - y.$$

If $g(y) < 0$ then $y_n < 0$ for large n , contradicting $y_n > 0$. Hence

$$(4) \quad f(y) \geq y \quad \text{for all } y > 0,$$

and moreover, since $y_{n+1} = f(y_n)$,

$$(5) \quad f(y + n g(y)) = y + (n + 1) g(y) \quad (n = 0, 1, 2, \dots).$$

Two elementary square-root estimates. (E1) For all real u with $-1 \leq u \leq 3$: $\sqrt{1+u} \geq 1 + \frac{u}{2} - \frac{u^2}{2}$.

Proof. If the right side is negative, this is trivial. Otherwise both sides are nonnegative and squaring is equivalent; expanding,

$$(1+u) - \left(1 + \frac{u}{2} - \frac{u^2}{2}\right)^2 = \frac{3}{4}u^2 + \frac{1}{2}u^3 - \frac{1}{4}u^4 = -\frac{u^2}{4}(u-3)(u+1) \geq 0$$

for $-1 \leq u \leq 3$. $\square$

(E2) For $A > 0$, $B \geq 0$: $\sqrt{A^2 + B} \leq A + \frac{B}{2A}$.

Proof. $(A + \frac{B}{2A})^2 = A^2 + B + \frac{B^2}{4A^2} \geq A^2 + B$, and both sides are positive. $\square$

Step 3: a point with gap $d > 0$ pins down all gaps at large arguments. Suppose $g(a) = d > 0$ for some $a > 0$. By (5),

$$f(a + nd) = a + (n + 1)d \quad (n \geq 0). \quad (6)$$

Claim. For every $x \geq a + d$,

$$d - \frac{d^2}{4x} \leq g(x) \leq d + \frac{d^2}{16x - 4d}. \quad (7)$$

Proof. The set $\{a + (n+1)d : n \geq 0\} = \{a+d, a+2d, \dots\}$ is an arithmetic progression with step d whose least element is $a+d \leq x$. Let $m = \lfloor (x-a)/d \rfloor$; then $m \geq 1$ and $a+md \leq x < a+(m+1)d$, and x lies within $d/2$ of one of the two endpoints $a+md$, $a+(m+1)d$, both of which belong to the set. Hence there is an index $n \geq 0$ with

$$|a + (n+1)d - x| \leq \frac{d}{2}.$$

Write $y := a + nd$ and $\varepsilon := (y+d) - x$, so $f(y) = y+d = x+\varepsilon$ by (6) and $|\varepsilon| \leq d/2$. Note $x \geq a+d > d$, hence $|\varepsilon|/x < \frac{1}{2}$.

Lower bound. By (2), $(f(x) + y)^2 \geq 4xf(y) = 4x(x+\varepsilon)$; since $f(x) + y > 0$,

$$f(x) \geq 2\sqrt{x(x+\varepsilon)} - y = 2x\sqrt{1 + \frac{\varepsilon}{x}} - (x + \varepsilon - d).$$

By (E1) with $u = \varepsilon/x \in (-\frac{1}{2}, \frac{1}{2})$,

$$2x\sqrt{1 + \frac{\varepsilon}{x}} \geq 2x\left(1 + \frac{\varepsilon}{2x} - \frac{\varepsilon^2}{2x^2}\right) = 2x + \varepsilon - \frac{\varepsilon^2}{x},$$

so

$$f(x) \geq 2x + \varepsilon - \frac{\varepsilon^2}{x} - x - \varepsilon + d = x + d - \frac{\varepsilon^2}{x} \geq x + d - \frac{d^2}{4x},$$

using $\varepsilon^2 \leq d^2/4$. This is the left inequality of (7).

Upper bound. By (1), $(f(x) + y)^2 \leq 2x^2 + 2f(y)^2 = 2x^2 + 2(x+\varepsilon)^2$. A direct expansion gives the identity

$$2x^2 + 2(x+\varepsilon)^2 = (2x+\varepsilon)^2 + \varepsilon^2.$$

Since $2x+\varepsilon \geq 2x - \frac{d}{2} > 0$, (E2) with $A = 2x+\varepsilon$, $B = \varepsilon^2$ yields

$$f(x) \leq \sqrt{(2x+\varepsilon)^2 + \varepsilon^2} - y \leq 2x+\varepsilon + \frac{\varepsilon^2}{2(2x+\varepsilon)} - (x+\varepsilon-d) = x+d + \frac{\varepsilon^2}{2(2x+\varepsilon)}.$$

Finally $\varepsilon^2 \leq d^2/4$ and $2(2x+\varepsilon) \geq 2(2x - \frac{d}{2}) = 4x-d > 0$, so

$$f(x) \leq x+d + \frac{d^2/4}{4x-d} = x+d + \frac{d^2}{16x-4d},$$

the right inequality of (7). $\square$

Step 4: all positive gaps are equal. Suppose $g(a) = d > 0$ and $g(a') = d' > 0$. For every $x \geq \max(a+d, a'+d')$, (7) applied to both points gives

$$d - \frac{d^2}{4x} \leq g(x) \leq d' + \frac{d'^2}{16x-4d'},$$

so that $d - d' \leq \frac{d^2}{4x} + \frac{d'^2}{16x-4d'}$ for all such x . The right-hand side tends to 0 as $x \rightarrow \infty$, whence $d \leq d'$; by symmetry $d' \leq d$. Hence $d = d'$:

(8) all positive values of g are equal.

Step 5: if some gap is positive, then $g \equiv d$ on a half-line. Assume $g(a) = d > 0$. For every $x \geq a + d$ we have $x > d > \frac{d}{4}$, so the lower bound in (7) gives

$$g(x) \geq d - \frac{d^2}{4x} > 0.$$

By (8), a positive gap must equal d ; hence

$$f(x) = x + d \quad \text{for all } x \geq T := a + d. \quad (9)$$

Step 6: downward bootstrap — $f(x) = x + d$ on all of $\mathbb{R}_{>0}$. **Claim.** If $f(t) = t + d$ for all $t \geq T$ (some $T > 0$, $d > 0$), then $f(y) = y + d$ for every $y > 0$ with $y \geq T - d$.

Proof. Let $y > 0$ with $y + d \geq T$, and put $x := y + d \geq T$, so $f(x) = x + d = y + 2d$. - Inequality (1) with this pair: $2(y+d)^2 + 2f(y)^2 \geq (y+2d+y)^2 = 4(y+d)^2$, hence $f(y)^2 \geq (y+d)^2$, i.e. $f(y) \geq y+d$. - Inequality (2) with this pair: $(y+2d+y)^2 \geq 4(y+d)f(y)$, i.e. $4(y+d)^2 \geq 4(y+d)f(y)$, hence $f(y) \leq y+d$.

Therefore $f(y) = y + d$. $\square$

Starting from (9) and applying the Claim repeatedly, we get by induction on $k \geq 0$: $f(y) = y + d$ for all $y > 0$ with $y \geq T - kd$. Indeed, the base case $k = 0$ is (9); for the inductive step, if $f(t) = t + d$ holds for all $t > 0$ with $t \geq T - kd$, then for any $y > 0$ with $y \geq T - (k+1)d$ the point $x = y + d$ satisfies $x > 0$ and $x \geq T - kd$, so $f(x) = x + d$ and the two displayed computations above give $f(y) = y + d$. Since $T - kd \rightarrow -\infty$, every $y > 0$ is covered:

$$f(y) = y + d \quad \text{for all } y > 0.$$

Step 7: conclusion. Either $g(y) = f(y) - y$ is identically 0, in which case $f(y) = y = y + 0$; or some gap is positive, in which case Steps 3–6 give $f(y) = y + d$ for all $y > 0$ with the constant $d > 0$. In both cases

$$f(x) = x + c \quad \text{for all } x > 0, \text{ for some constant } c \geq 0.$$

Together with Part I, the set of solutions is exactly $\{x \mapsto x + c : c \geq 0\}$. $\blacksquare$

IMO 2026 — Problem 6

Solution by Claude Fable 5

2026-07-18

Source: `results/claude-fable-5/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Approaches tried

- **Simulation / pattern finding** (`code/verify.py`): for many seeds a_1 the set $S = \{a_n\}$ is exactly $[a_1, \infty) \cap \bigcup_Q d_Q \mathbb{Z}$ for a finite family of squarefree moduli d_Q (e.g. $a_1 = 15$: moduli 6, 10, 15; $a_1 = 385$: moduli 30, 66, 14, 418, 231, 399, 385), and $a_{n+T} = a_n + L$ holds from $n = 1$. This suggested the exact structural characterization.
- **Self-consistency + blocker (hypergraph transversal) reduction**: proved $S = \{m \geq a_1 : \gcd(m, s) > 1 \forall s \in S\}$ and hence $S = [a_1, \infty) \cap \bigcup_{Q \in \mathcal{M}} d_Q \mathbb{Z}$ where $\mathcal{M}$ is the set of minimal transversals of the family of prime-divisor sets of elements of S . Reduced the problem to: $\mathcal{M}$ is finite.

- **Dead end — pure set-system arguments** (approaches/abstract-clutter-dead-end.md): tried to prove finiteness of $\mathcal{M}$ from “self-blocking clutter” axioms alone (pairwise intersecting antichain in which every finite transversal contains a member). Found an explicit **infinite** self-blocking clutter of finite sets $(U_k(u) = \{u_1, \dots, u_k, x_{k+1}, y_{k+1}\}, u_i \in \{x_i, y_i\})$, so the greedy (“smallest next term”) property must be used quantitatively, not just the fixed-point property of S .
- **Successful approach** (approaches/main-blocker-descent.md): from greediness, every non-element of S has a *smaller* coprime witness in S . Applied to $m = \pi(Q \setminus \{t\})^e$ this yields a “cheap witness” member Q' with $Q' \cap Q = \{t\}$ and $d_{Q'} < a_1 d_Q / t$. Iterating at a fixed prime $t = q > a_1$ gives a geometric descent $d \mapsto (a_1/q)d$ of member products, an impossibility; hence all primes occurring in members are $\leq a_1$, so $\mathcal{M}$ is finite and S is a finite union of arithmetic progressions from a_1 on. Periodicity (from $n = 1!$) follows at once.

Current best

Complete proof. With $\mathcal{M}$ the (finite, as proved) set of minimal transversals of $\{P(s) : s \in S\}$, one may take $L = M := \prod_{p \leq a_1, p \text{ prime}} p$ and $T = \#\{r \in \{0, \dots, M-1\} : \exists Q \in \mathcal{M}, d_Q \mid r\} (\geq 1)$; then $a_{n+T} = a_n + L$ for **every** $n \geq 1$. Verified numerically for ~ 70 seeds (all $2 \leq a_1 \leq 60$ and assorted larger ones, ranges up to $4 \cdot 10^5$).

Full proof

Throughout, “prime set” means a finite set of prime numbers. For an integer $n > 1$ let $P(n)$ denote the set of primes dividing n ; it is finite and nonempty. For a prime set X put

$$\pi(X) := \prod_{p \in X} p \quad (\pi(\emptyset) = 1).$$

Note that for an integer $m \geq 1$ and a prime set X ,

$$X \subseteq P(m) \iff \pi(X) \mid m, \quad (0)$$

since $\pi(X)$ is a product of distinct primes.

By its definition the sequence is strictly increasing, so all terms are distinct and $a_n \rightarrow \infty$. Let

$$S := \{a_1, a_2, a_3, \dots\}.$$

Every element of S is $\geq a_1 \geq 2$.

Step 1: Basic properties of S

Lemma 1 (pairwise non-coprimality). For all $s, s' \in S$ we have $\gcd(s, s') > 1$.

Proof. If $s = s'$ then $\gcd(s, s') = s > 1$. If $s = a_i, s' = a_j$ with $i < j$, then $j \geq 2$ and, by the defining property of a_j (applied with $n = j - 1$), $\gcd(a_j, a_i) > 1$. $\square$

Lemma 2 (completeness). $S = \{m \in \mathbb{Z} : m \geq a_1 \text{ and } \gcd(m, s) > 1 \text{ for all } s \in S\}$.

Proof. “ $\subseteq$ ”: every element of S is $\geq a_1$, and Lemma 1 applies.

“ $\supseteq$ ”: Let $m \geq a_1$ satisfy $\gcd(m, s) > 1$ for all $s \in S$, and suppose $m \notin S$. Then $m > a_1$. Since $a_n \rightarrow \infty$ and $a_1 < m$, the index

$$n := \max\{i \geq 1 : a_i < m\}$$

is well defined. By maximality of n we have $a_{n+1} \geq m$, and $a_{n+1} \neq m$ because $m \notin S$; hence $a_{n+1} > m$. On the other hand $m > a_n$ and $\gcd(m, a_i) > 1$ for all $i \leq n$ (these a_i lie in S), so m is an integer greater than a_n having a nontrivial common factor with each of $a_1, \dots, a_n$. By the minimality in the definition of a_{n+1} , $a_{n+1} \leq m$ — a contradiction. $\square$

Lemma 3 (downward witnesses; this is where greediness enters). If $m > a_1$ and $m \notin S$, then there exists $s \in S$ with $s < m$ and $\gcd(s, m) = 1$.

Proof. Define $n := \max\{i \geq 1 : a_i < m\}$ as in Lemma 2; as there, $a_n < m < a_{n+1}$. If we had $\gcd(m, a_i) > 1$ for every $i \leq n$, then m would be a candidate in the definition of a_{n+1} , forcing $a_{n+1} \leq m$, a contradiction. Hence $\gcd(m, a_i) = 1$ for some $i \leq n$, and $s := a_i \leq a_n < m$ works. $\square$

Step 2: Transversals and the structure of S

Definition. A prime set X is a *transversal* if $X \cap P(s) \neq \emptyset$ for every $s \in S$. A *member* is a transversal Q such that no proper subset of Q is a transversal. Let $\mathcal{M}$ denote the set of all members, and for $Q \in \mathcal{M}$ put $d_Q := \pi(Q)$.

Two immediate remarks:

- $\emptyset$ is not a transversal (as $S \neq \emptyset$ and each $P(s)$ is disjoint from $\emptyset$); hence every member is nonempty, so $d_Q \geq 2$.
- **Every transversal X contains a member.** Indeed, among the finitely many subsets of X that are transversals (a nonempty collection, containing X), choose one, Q , that is inclusion-minimal in this collection. If some proper subset $Q'' \subsetneq Q$ were a transversal, then $Q'' \subseteq X$ would contradict the minimal choice of Q . So Q is a member. (*)

Lemma 4 (exact description of S). $\mathcal{M} \neq \emptyset$, and for every integer $m \geq a_1$:

$$m \in S \iff d_Q \mid m \text{ for some } Q \in \mathcal{M}.$$

Proof. “ $\Leftarrow$ ”: Suppose $d_Q \mid m$, i.e. (by (0)) $Q \subseteq P(m)$. For every $s \in S$, since Q is a transversal, $\emptyset \neq Q \cap P(s) \subseteq P(m) \cap P(s)$, i.e. $\gcd(m, s) > 1$. As $m \geq a_1$, Lemma 2 gives $m \in S$.

“ $\Rightarrow$ ”: Let $m \in S$. By Lemma 1, $\gcd(m, s) > 1$ for every $s \in S$, i.e. $P(m)$ is a transversal. By (*) it contains a member Q , and $Q \subseteq P(m)$ means $d_Q \mid m$ by (0).

Finally, $P(a_1)$ is a transversal (Lemma 1), so it contains a member; thus $\mathcal{M} \neq \emptyset$. $\square$

Lemma 5 (members pairwise intersect). For all $Q, Q' \in \mathcal{M}$ we have $Q \cap Q' \neq \emptyset$.

Proof. Choose distinct primes $p, p' \notin Q \cup Q'$ (possible: there are infinitely many primes). Choose integers $e, e' \geq 1$ with $m := d_Q p^e \geq a_1$ and $m' := d_{Q'} p'^{e'} \geq a_1$. By Lemma 4 (“ $\Leftarrow$ ”), $m, m' \in S$. By Lemma 1, $\gcd(m, m') > 1$, so some prime u lies in

$$P(m) \cap P(m') = (Q \cup \{p\}) \cap (Q' \cup \{p'\}).$$

Since $p \neq p'$, $p \notin Q'$ and $p' \notin Q$, necessarily $u \in Q \cap Q'$. $\square$

Step 3: The key lemma — cheap witnesses

Lemma 6 (cheap witness). Let $Q \in \mathcal{M}$ with $|Q| \geq 2$, and let $t \in Q$. Then there exists $Q' \in \mathcal{M}$ with

$$Q' \cap Q = \{t\} \quad \text{and} \quad d_{Q'} < \frac{a_1 d_Q}{t}.$$

Proof. Put $X := Q \setminus \{t\}$; then $X \neq \emptyset$ and $\pi(X) = d_Q/t \geq 2$.

First, **no member is a subset of X** : if $M \in \mathcal{M}$ and $M \subseteq X \subsetneq Q$, then M would be a transversal that is a proper subset of the member Q , contradicting the definition of a member.

Let $e \geq 1$ be minimal with $m := \pi(X)^e > a_1$. Then

$$m \leq a_1 \pi(X) : \quad (1)$$

if $e = 1$ this is clear since $m = \pi(X) \leq a_1 \pi(X)$; if $e \geq 2$, minimality of e gives $\pi(X)^{e-1} \leq a_1$, so $m = \pi(X)^{e-1} \cdot \pi(X) \leq a_1 \pi(X)$.

Since $P(m) = X$ contains no member and $m > a_1$, Lemma 4 gives $m \notin S$. By Lemma 3 there exists $s \in S$ with $s < m$ and $\gcd(s, m) = 1$, i.e.

$$P(s) \cap X = \emptyset. \quad (2)$$

By Lemma 4 (“ $\Rightarrow$ ”) there is a member $Q' \subseteq P(s)$. Then by (2), $Q' \cap X = \emptyset$, and by (0), $d_{Q'} \mid s$, hence using (1):

$$d_{Q'} \leq s < m \leq a_1 \pi(X) = \frac{a_1 d_Q}{t}.$$

By Lemma 5, $Q' \cap Q \neq \emptyset$; since $Q = X \cup \{t\}$ and $Q' \cap X = \emptyset$, this forces $Q' \cap Q = \{t\}$. $\square$

Step 4: Members contain no prime exceeding a_1

Lemma 7. Every prime belonging to some member is $\leq a_1$.

Proof. **Case A: some member is a singleton $\{p\}$.** Let $M \in \mathcal{M}$ be arbitrary. By Lemma 5, $M \cap \{p\} \neq \emptyset$, i.e. $p \in M$. Then $\{p\} \subseteq M$ is a transversal, and since no proper subset of the member M is a transversal, $M = \{p\}$. Hence $\mathcal{M} = \{\{p\}\}$. Applying Lemma 4 (“ $\Rightarrow$ ”) to $a_1 \in S$ gives $p \mid a_1$, so $p \leq a_1$, and the lemma holds in this case.

Case B: every member has at least two elements. Suppose, for contradiction, that some member Q_0 contains a prime $q > a_1$. Define recursively, for $i \geq 0$: $Q_{i+1} :=$ the member provided by Lemma 6 applied to $Q := Q_i$ and $t := q$. This is legitimate by induction: $q \in Q_i$ (for $i = 0$ by assumption; for $i \geq 1$ because $Q_i \cap Q_{i-1} = \{q\}$), and $|Q_i| \geq 2$ by the Case B hypothesis. Lemma 6 yields

$$d_{Q_{i+1}} < \frac{a_1}{q} d_{Q_i} \quad (i \geq 0), \quad \text{hence} \quad d_{Q_i} < \left(\frac{a_1}{q}\right)^i d_{Q_0} \quad (i \geq 1).$$

Since $q > a_1$, we have $a_1/q < 1$, so $d_{Q_i} \rightarrow 0$ as $i \rightarrow \infty$. But $q \in Q_i$ implies $d_{Q_i} \geq q \geq 2$ for all i — a contradiction. $\square$

Lemma 8 (finiteness). $\mathcal{M}$ is finite and nonempty. Moreover, with

$$M := \prod_{\substack{p \text{ prime} \\ p \leq a_1}} p \quad (\geq 2, \text{ since } a_1 \geq 2),$$

every $Q \in \mathcal{M}$ satisfies $d_Q \mid M$.

Proof. By Lemma 7, every member is a subset of the finite set $\Pi := \{p \text{ prime} : p \leq a_1\}$; hence $\mathcal{M} \subseteq 2^\Pi$ is finite. It is nonempty by Lemma 4. Each d_Q is a product of distinct primes $\leq a_1$, hence divides M . $\square$

Step 5: Conclusion — exact linear periodicity

Define

$$R := \{r \in \{0, 1, \dots, M-1\} : d_Q \mid r \text{ for some } Q \in \mathcal{M}\}, \quad T := |R|, \quad L := M.$$

Since $d_Q \mid 0$ for any $Q \in \mathcal{M} \neq \emptyset$, we have $0 \in R$, so $T \geq 1$; and $L = M \geq 2$. Thus T, L are positive integers.

Because $d_Q \mid M$ for all $Q \in \mathcal{M}$ (Lemma 8), for any integer m we have $d_Q \mid m \iff d_Q \mid (m \bmod M)$. Hence Lemma 4 can be restated: for every integer $m \geq a_1$,

$$m \in S \iff (m \bmod M) \in R. \quad (3)$$

In particular,

$$\text{for every integer } m \geq a_1 : \quad m \in S \iff m + M \in S. \quad (4)$$

For $k \geq 0$ let $I_k := \{a_1 + kM, a_1 + kM + 1, \dots, a_1 + (k+1)M - 1\}$, a block of M consecutive integers, all $\geq a_1$. Each I_k contains exactly one integer from each residue class modulo M , so by (3),

$$|S \cap I_k| = |R| = T \quad \text{for every } k \geq 0. \quad (5)$$

Also $S \subseteq \bigcup_{k \geq 0} I_k$, because $\min S = a_1$.

Claim. For every $k \geq 0$, the elements of S smaller than $a_1 + kM$ are exactly $a_1, a_2, \dots, a_{kT}$ (so $S \cap I_k = \{a_{kT+1} < a_{kT+2} < \dots < a_{(k+1)T}\}$).

Proof of Claim. Induction on k . For $k = 0$ the statement is vacuously true (S has no elements below a_1). Assume it for k . The elements of S smaller than $a_1 + (k+1)M$ are those smaller than $a_1 + kM$ — namely $a_1, \dots, a_{kT}$ — together with $S \cap I_k$, which by (5) has exactly T elements, each larger than every element of S below $a_1 + kM$. Since $a_1 < a_2 < \dots$ enumerates S in increasing order, $S \cap I_k = \{a_{kT+1}, \dots, a_{(k+1)T}\}$ and the elements below $a_1 + (k+1)M$ are $a_1, \dots, a_{(k+1)T}$. $\square$

By (4), the map $\varphi(m) := m + M$ maps $S \cap I_k$ into $S \cap I_{k+1}$ (if $m \in S \cap I_k$ then $m \geq a_1$, so $m + M \in S$, and clearly $m + M \in I_{k+1}$), and the map $m' \mapsto m' - M$ maps $S \cap I_{k+1}$ into $S \cap I_k$ (if $m' \in S \cap I_{k+1}$ then $m' - M \geq a_1$ and $m' - M \in S$ by (4) applied to $m := m' - M$). These maps are mutually inverse, so $\varphi : S \cap I_k \rightarrow S \cap I_{k+1}$ is a strictly increasing bijection between two T -element sets. A strictly increasing bijection between finite sets listed in increasing order matches the j -th smallest element to the j -th smallest element; combined with the Claim this gives

$$a_{(k+1)T+j} = a_{kT+j} + M \quad \text{for all } k \geq 0, 1 \leq j \leq T.$$

Finally, let $n \geq 1$ be arbitrary and write $n = kT + j$ with $k \geq 0$ and $1 \leq j \leq T$. Then

$$a_{n+T} = a_{(k+1)T+j} = a_{kT+j} + M = a_n + L.$$

Thus $a_{n+T} = a_n + L$ for every positive integer n , with the positive integers T and L defined above.

■

Remarks

- The proof shows more: $S = \{m \geq a_1 : d_Q \mid m \text{ for some member } Q\}$ is *exactly* a finite union of arithmetic progressions intersected with $[a_1, \infty)$ — the periodic description is valid from the very first term, which is why no “eventual” qualifier is needed.
- Greediness is used exactly once, in Lemma 3 (every excluded integer has a **smaller** coprime witness in S); this is essential: there exist “self-consistent” sets S (satisfying Lemma 2 but not Lemma 3) built from an infinite self-blocking clutter which are not unions of finitely many progressions. See `approaches/abstract-clutter-dead-end.md`.
- Numerical verification: `code/verify.py` confirms Lemmas 4, 5, 7 and the Theorem (with these exact T, L up to replacing M by $\text{lcm}_Q d_Q$, a divisor of M ; both work) for all $2 \leq a_1 \leq 60$ and for $a_1 \in \{77, 91, 101, 105, 121, 125, 143, 169, 255, 385, 663, 1001, 1155\}$, over ranges up to $4 \cdot 10^5$.